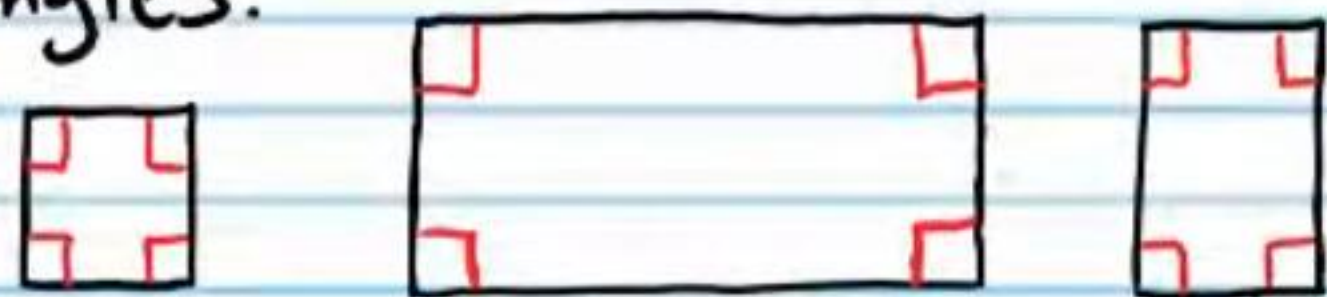


Special Quadrilaterals (Part II): Kites, Rhombi, Trapezoids, Rectangles, and Squares

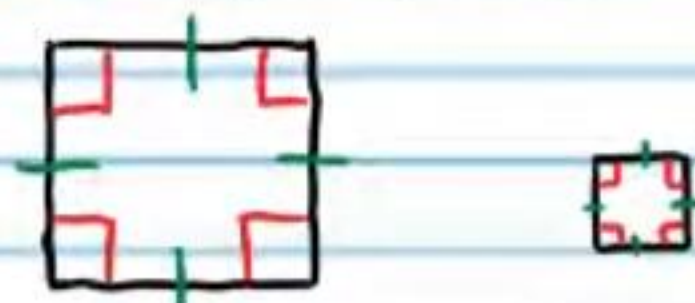
Rectangle: A quadrilateral whose inner angles are all right angles.

Examples:



Square: A quadrilateral whose inner angles are all right angles and whose sides are all congruent.

Examples:



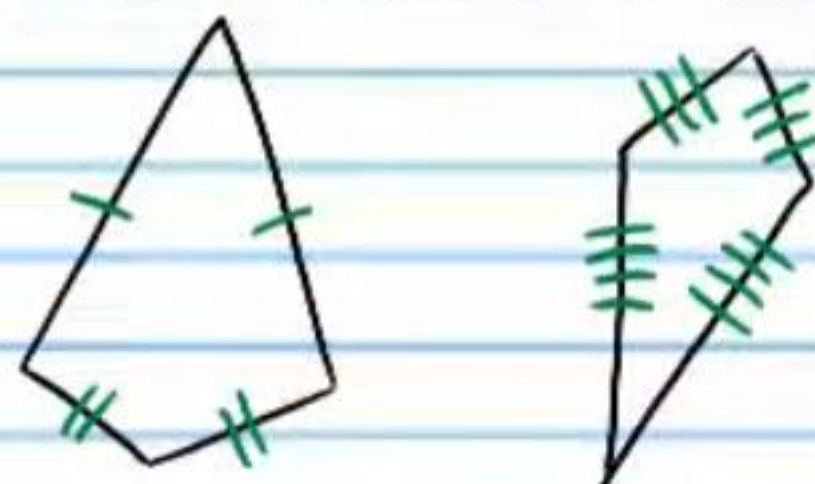
Rhombus: A quadrilateral whose sides are all congruent.

Examples:



Kite: A quadrilateral with exactly two pairs of congruent consecutive sides.

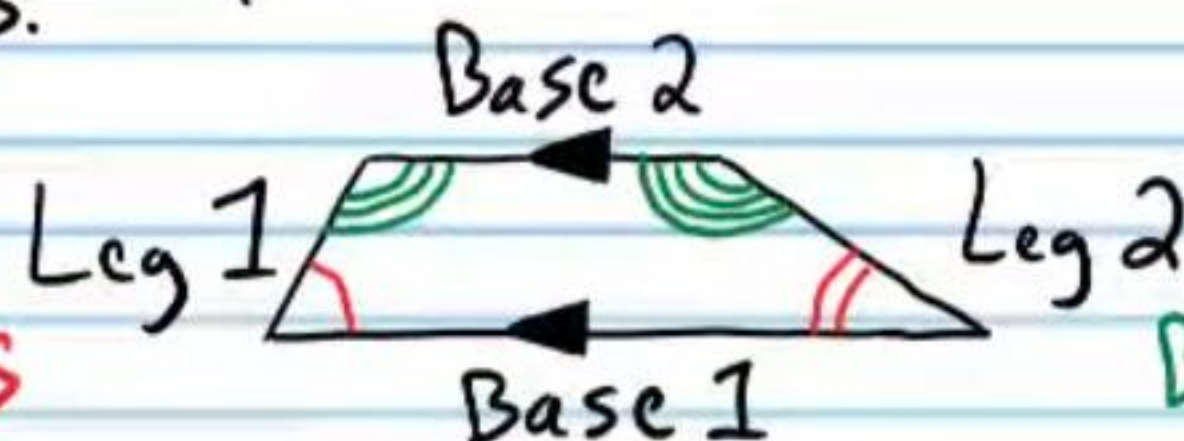
Examples:



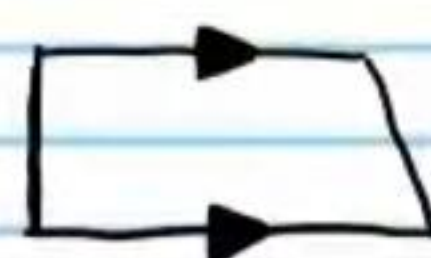
Trapezoid: A quadrilateral with exactly one pair of parallel sides.

Examples:

Base Angles

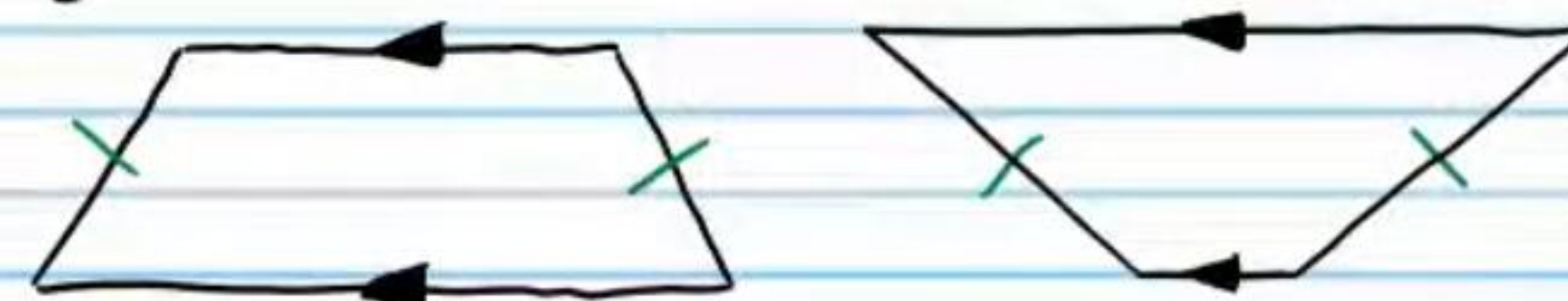


Base Angles

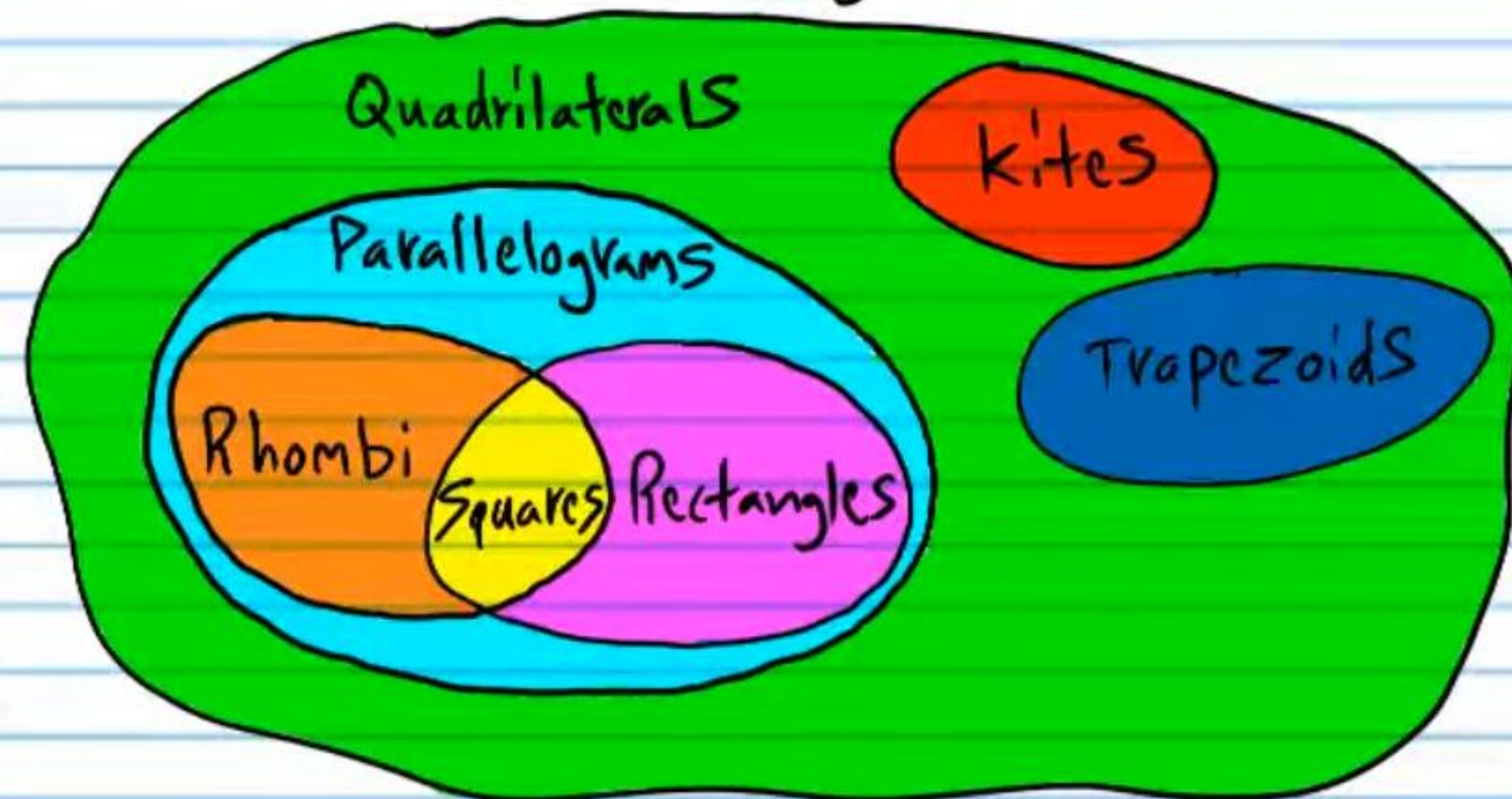


Isosceles Trapezoid: A trapezoid with congruent legs.

Examples:



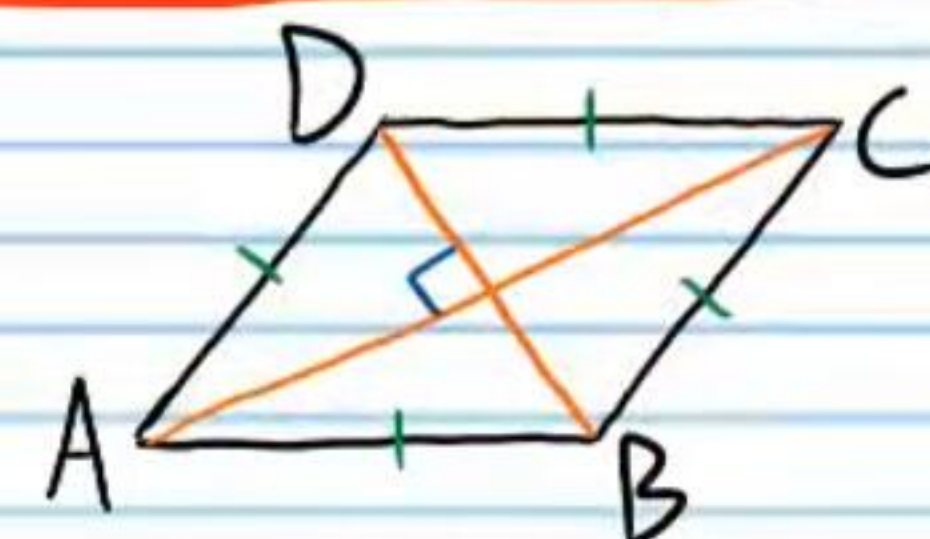
Venn Diagram



Rhombi

Rhombus Perpendicular Diagonals Theorem: If a quadrilateral is a rhombus, then its diagonals are perpendicular.

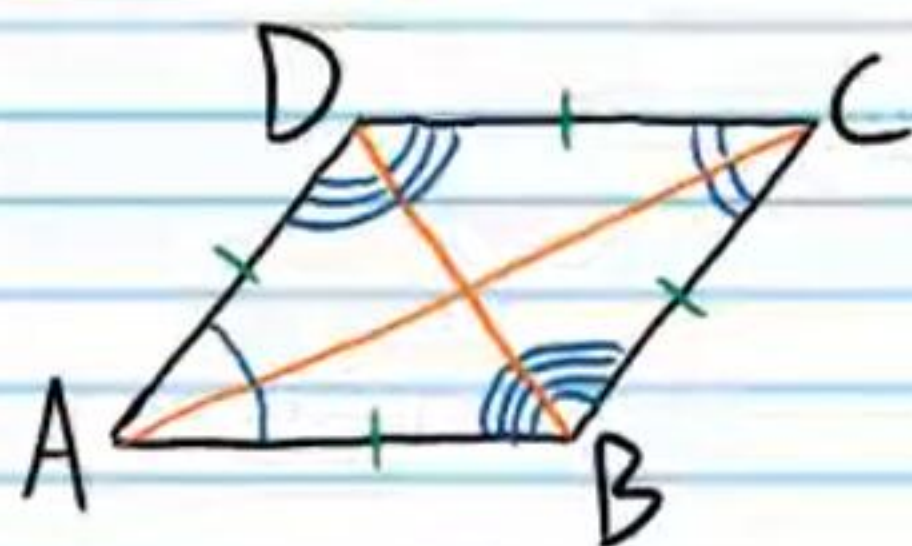
Example:



$\overline{AC} \perp \overline{DB}$

Rhombus Angle Bisector Theorem: If a quadrilateral is a rhombus, then its diagonals bisect the inner angles.

Example:

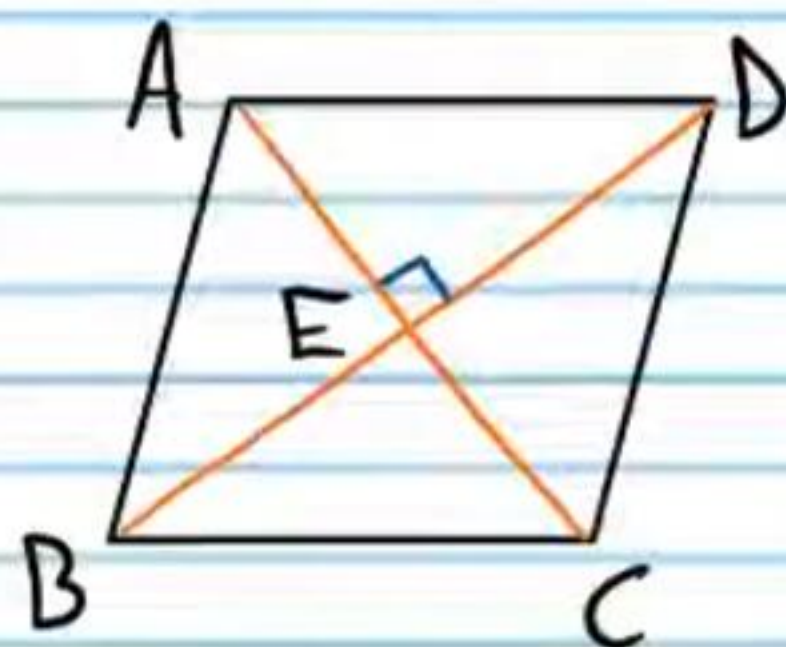


$$\begin{aligned}\angle DAC &\cong \angle CAB \\ \angle DCA &\cong \angle BCA \\ \angle ADB &\cong \angle CDB \\ \angle ABD &\cong \angle CBD\end{aligned}$$

Rhombus Parallelogram Theorem: If a quadrilateral is a rhombus, then it is a parallelogram.

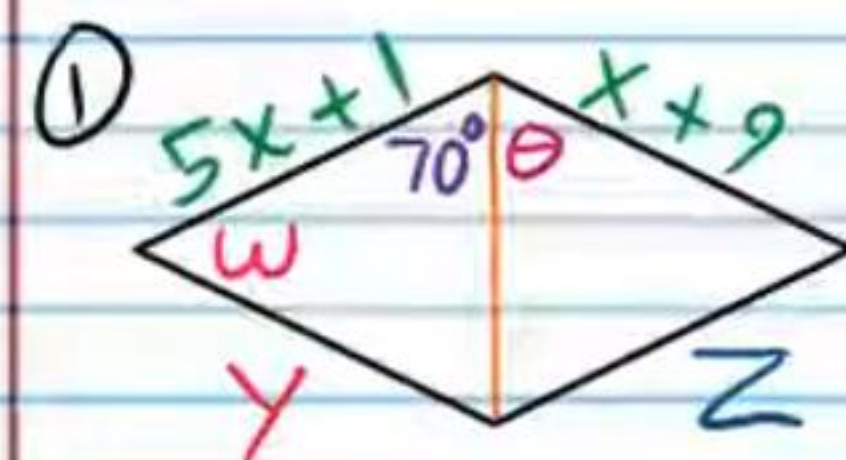
Rhombus Right Triangles Theorem: The diagonals of a rhombus divide it into four congruent right triangles.

Example:



$$\triangle ABE \cong \triangle ADE \cong \triangle CDE \cong \triangle CBE$$

Find the unknowns in each rhombus below.



$$\text{I) } 5x+1 = x+9$$

$$4x = 8$$

$$\boxed{x = 2}$$

$$\text{II) } y = 5(2)+1 = 11$$

$$\boxed{y = 11}$$

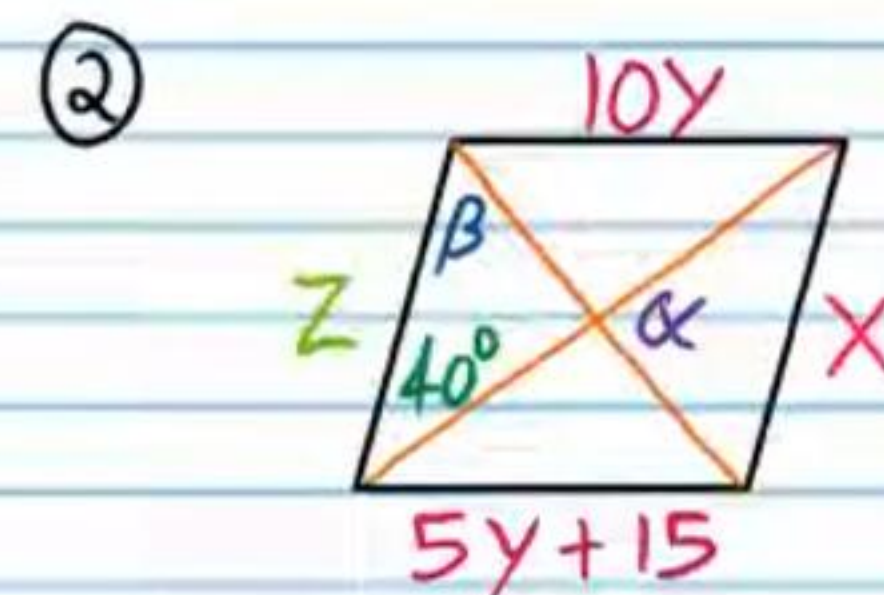
$$\text{III) } \boxed{z = 11}$$

$$\text{IV) } \boxed{\theta = 70^\circ}$$

$$\text{V) } w + 70 + 70 = 180$$

$$w + 140 = 180$$

$$\boxed{w = 40^\circ}$$



$$\text{I) } 10y = 5y+15$$

$$5y = 15$$

$$\boxed{y = 3}$$

$$\text{II) } z = 10(3) = 30$$

$$\boxed{z = 30}$$

$$\text{III) } \boxed{x = 30}$$

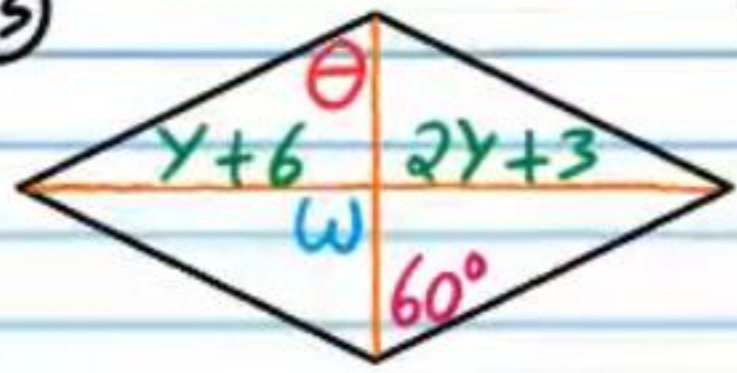
$$\text{IV) } \boxed{\alpha = 90^\circ}$$

$$\text{V) } \triangle \quad 40 + \beta + 90 = 180$$

$$\beta + 130 = 180$$

$$\boxed{\beta = 50^\circ}$$

③



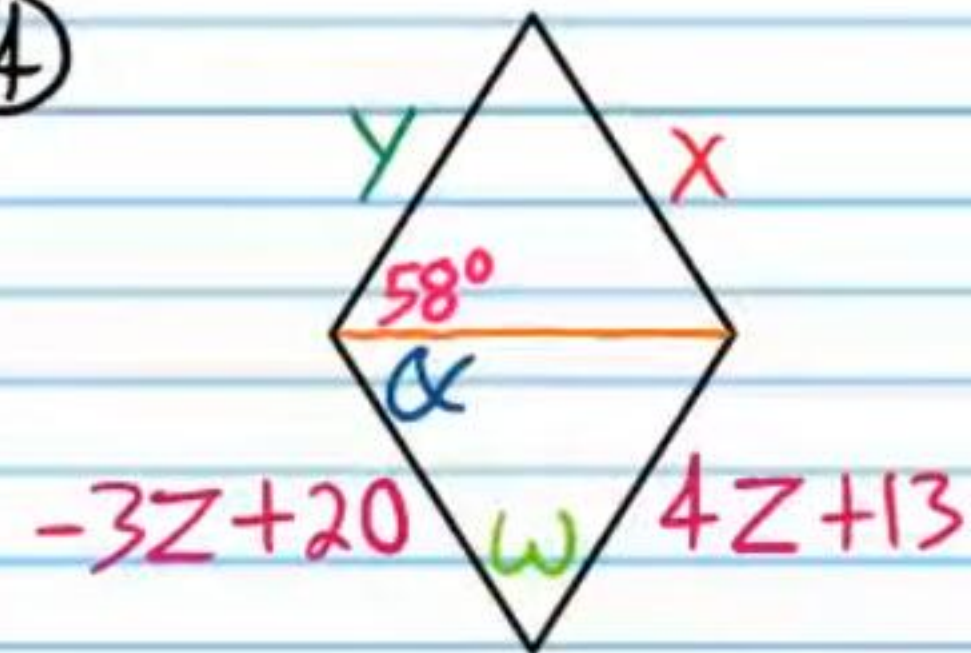
$$\text{I) } y+6 = 2y+3$$

$$\boxed{3 = y}$$

$$\text{II) } \boxed{w = 90^\circ}$$

$$\text{III) } \boxed{\theta = 60^\circ}$$

④



$$\text{I) } -3z+20 = 4z+13$$

$$7 = 7z$$

$$\boxed{1 = z}$$

$$\text{II) } y = -3(1)+20 = 17$$

$$\boxed{y = 17}$$

$$\text{III) } \boxed{x = 17}$$

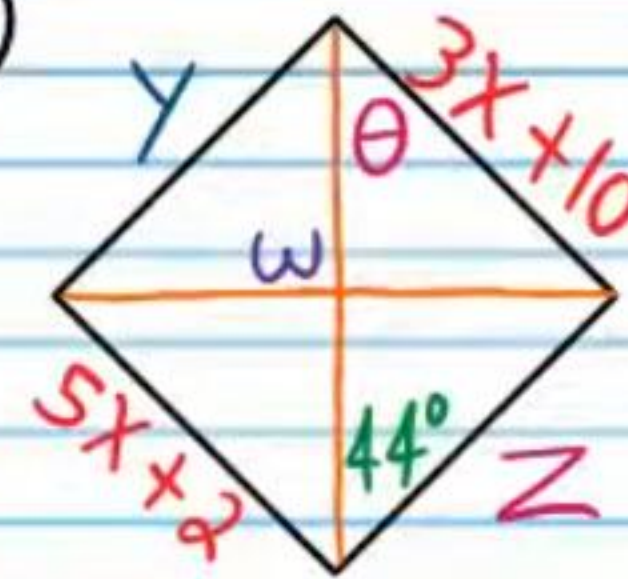
$$\text{IV) } \boxed{\theta = 58^\circ}$$

$$\text{V) } w + 58 + 58 = 180$$

$$w + 116 = 180$$

$$\boxed{w = 64^\circ}$$

⑤



$$\text{I) } \boxed{w = 90^\circ}$$

$$\text{II) } 3x+10 = 5x+2$$

$$8 = 2x$$

$$\boxed{4 = x}$$

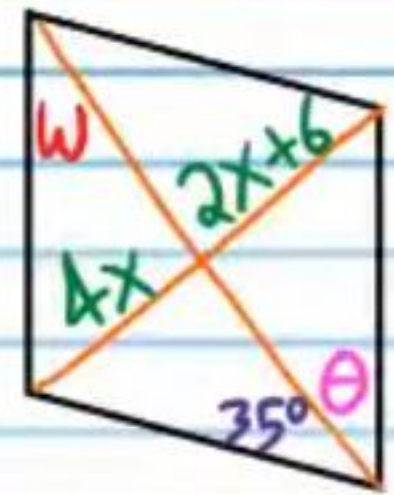
$$\text{III) } y = 3(4)+10 = 22$$

$$\boxed{y = 22}$$

$$\text{IV) } \boxed{z = 22}$$

$$\text{V) } \boxed{\theta = 44^\circ}$$

⑥



$$\text{I) } \boxed{\theta = 35^\circ}$$

$$\text{II) } \boxed{w = 35^\circ}$$

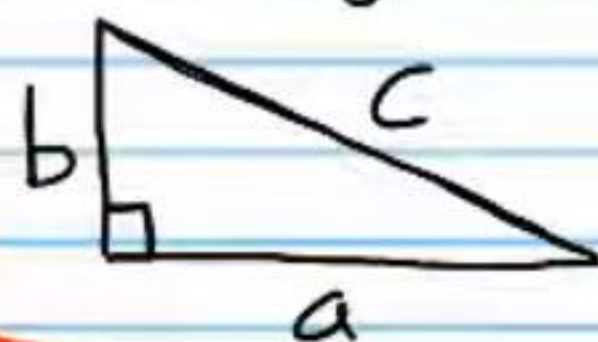
$$\text{III) } 4x = 2x+6$$

$$2x = 6$$

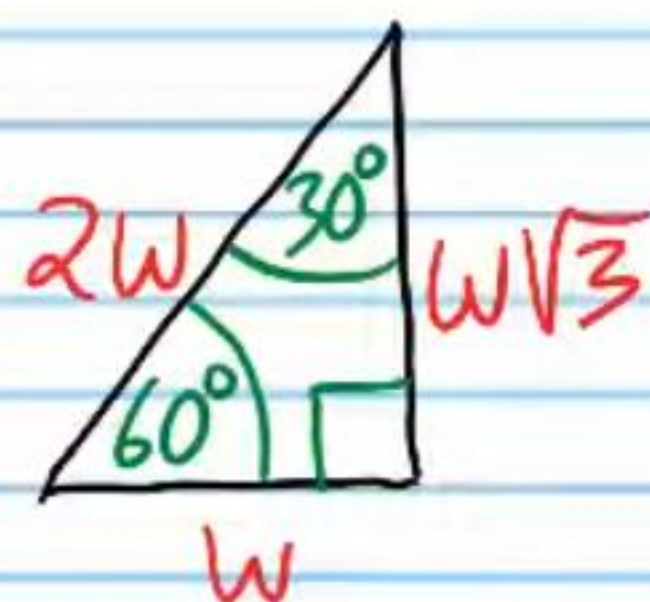
$$\boxed{x = 3}$$

Review

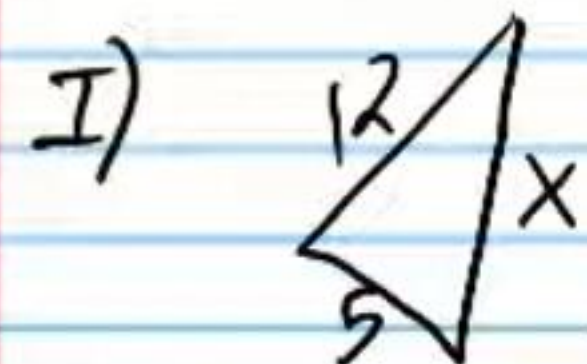
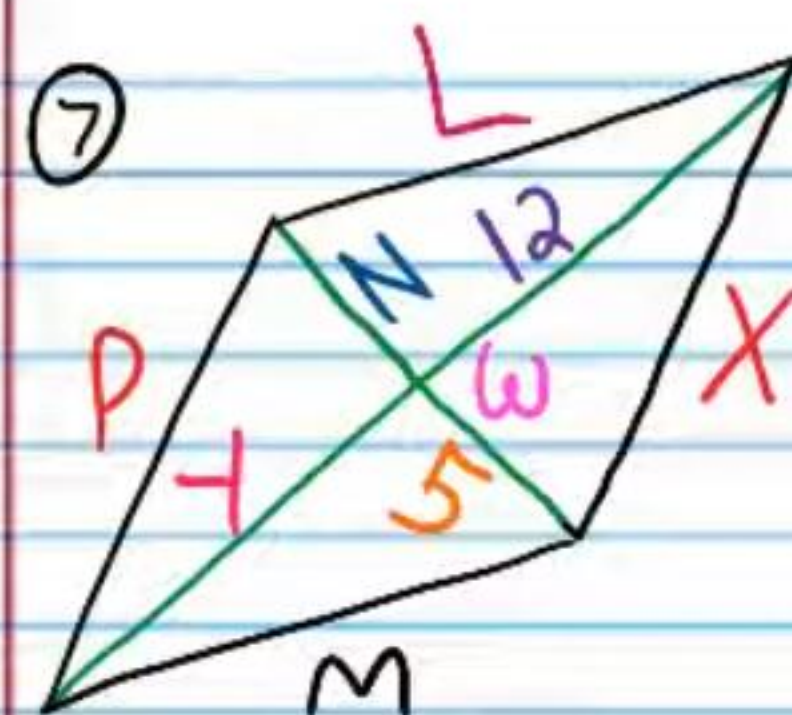
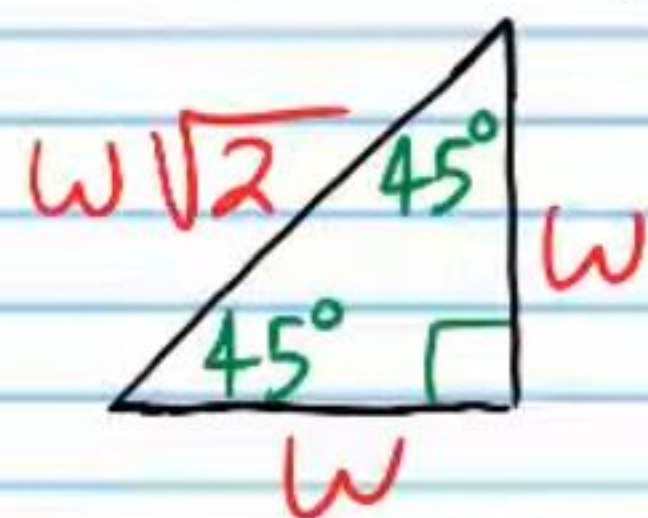
The Pythagorean Theorem: If a , b , and c are the legs and hypotenuse of a right triangle, respectively, then $a^2 + b^2 = c^2$.



30°-60°-90° Triangle Theorem: If the inner angles of a triangle are 30°, 60°, and 90°, and the length of the side opposite the 30° angle is w , then the length of the hypotenuse is $2w$ and the length of the side opposite the 60° angle is $w\sqrt{3}$.



45°-45°-90° Triangle Theorem: If the inner angles of a triangle are 45°, 45°, and 90°, and the length of the two sides opposite the 45° angles are w , then the length of the hypotenuse is $w\sqrt{2}$.



$$a^2 + b^2 = c^2$$

$$12^2 + 5^2 = x^2$$

$$144 + 25 = x^2$$

$$169 = x^2$$

$$\pm\sqrt{169} = \sqrt{x^2}$$

$$\pm 13 = x$$

$$13 = x$$

$$\text{II) } \boxed{P = 13}$$

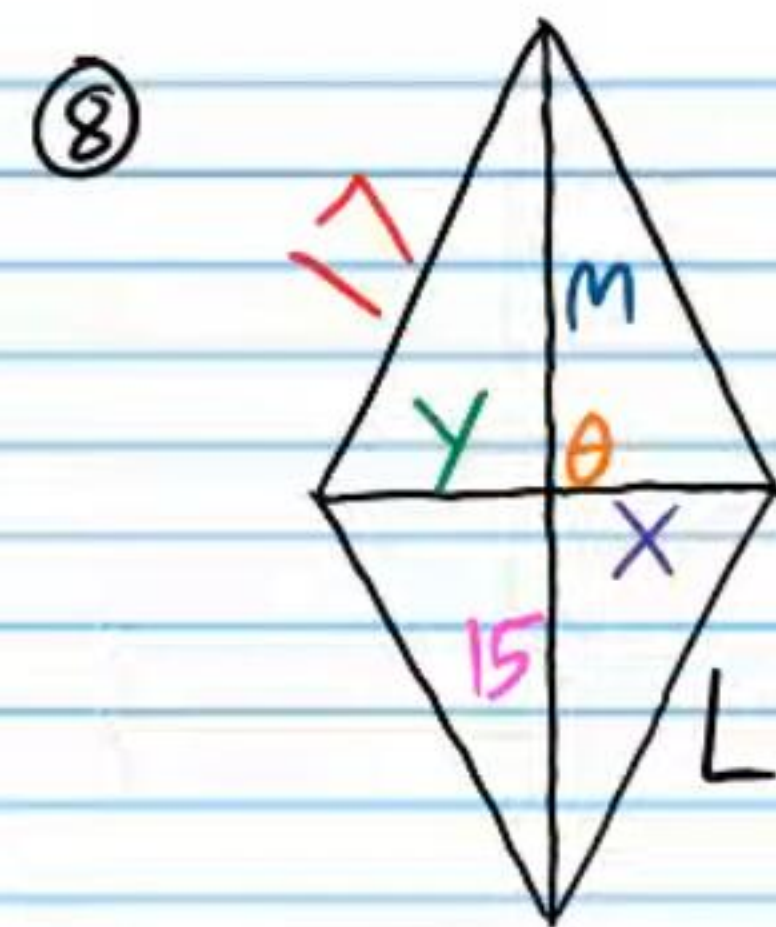
$$\text{III) } \boxed{L = 13}$$

$$\text{IV) } \boxed{W = 90^\circ}$$

$$\text{V) } \boxed{Y = 12}$$

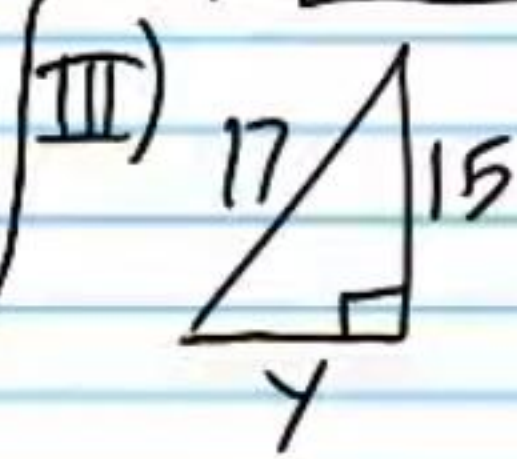
$$\text{VI) } \boxed{Z = 5}$$

$$\text{VII) } \boxed{M = 13}$$



$$\text{I) } \boxed{\theta = 90^\circ}$$

$$\text{II) } \boxed{M = 15}$$



$$a^2 + b^2 = c^2$$

$$15^2 + y^2 = 17^2$$

$$225 + y^2 = 289$$

$$y^2 = 64$$

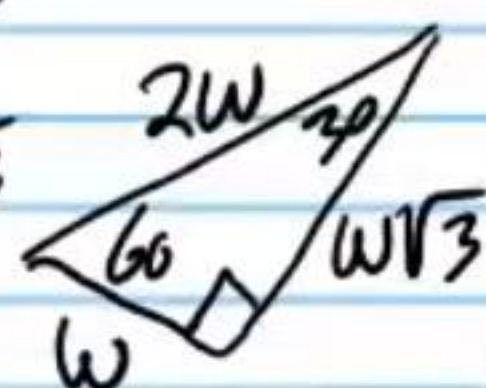
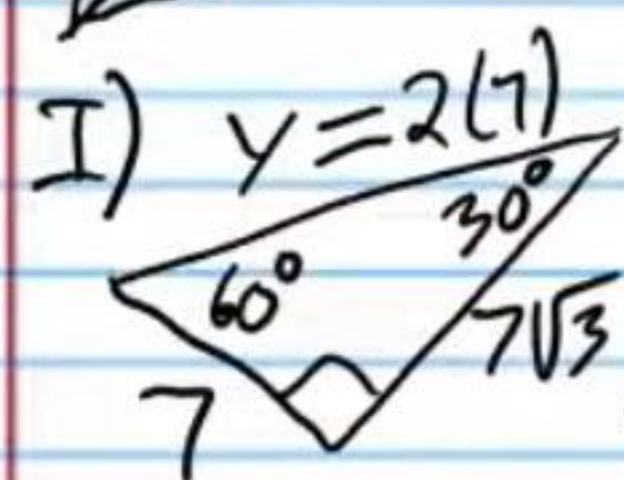
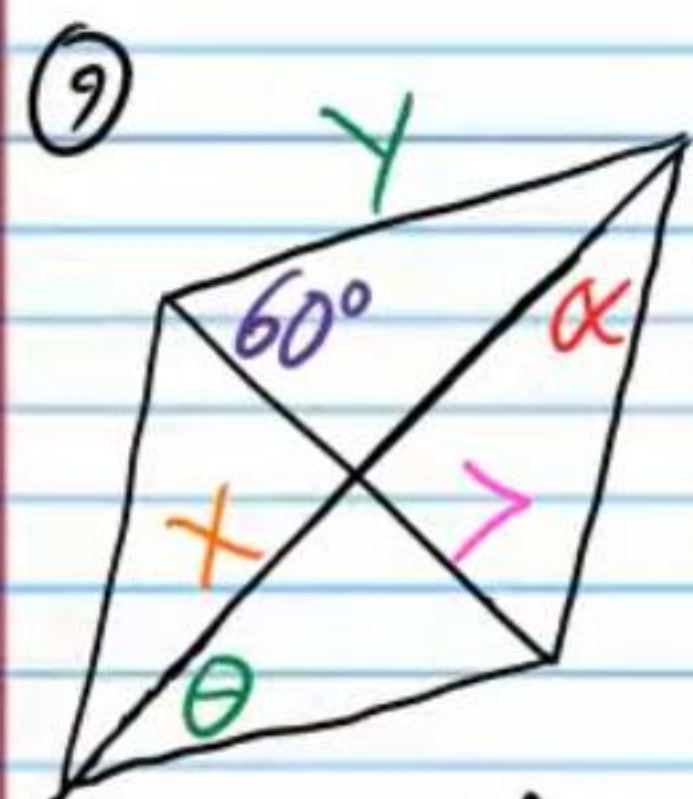
$$\sqrt{y^2} = \pm\sqrt{64}$$

$$y = \pm 8$$

$$\boxed{y = 8}$$

$$\text{IV) } \boxed{x = 8}$$

$$\text{V) } \boxed{L = 17}$$



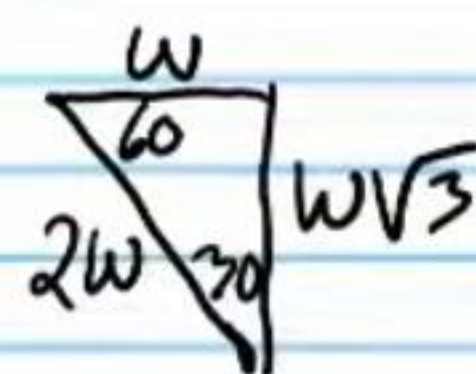
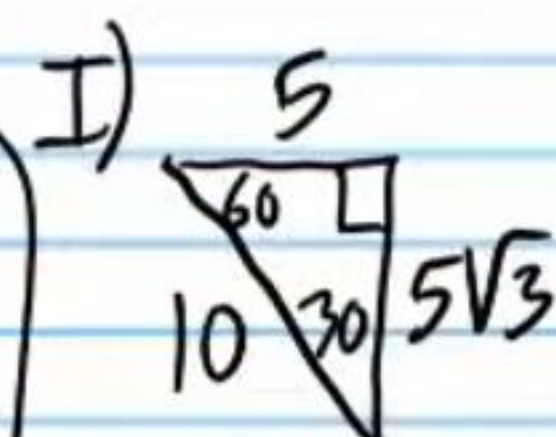
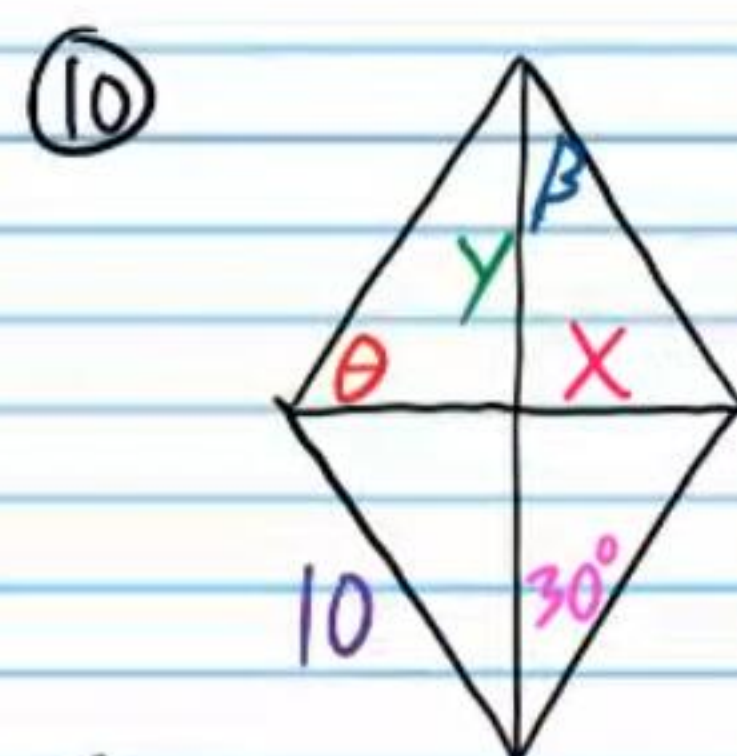
$$y = 2(7) = 14$$

$$\boxed{y = 14}$$

II) $\boxed{x = 7\sqrt{3}}$

III) $\boxed{\alpha = 30^\circ}$

IV) $\boxed{\theta = 30^\circ}$



II) $\boxed{y = 5\sqrt{3}}$

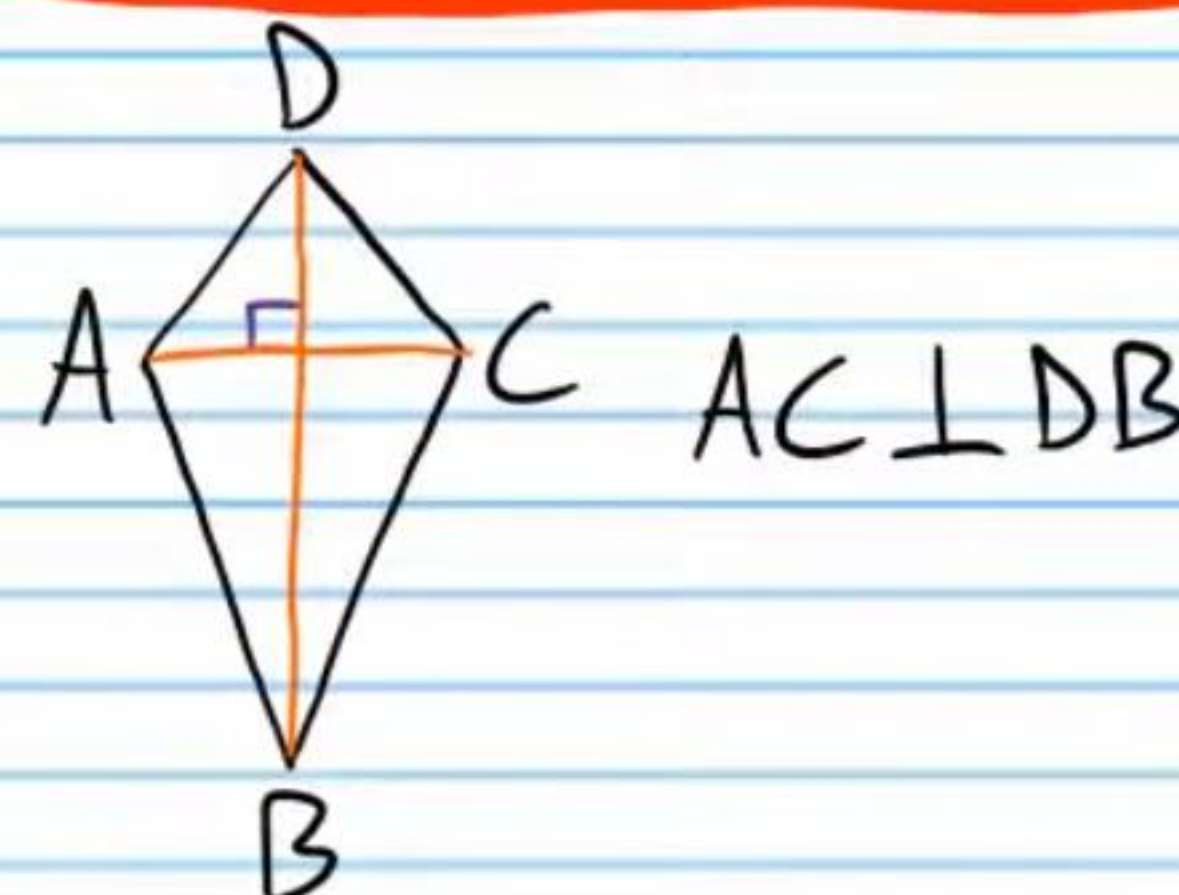
III) $\boxed{\beta = 30^\circ}$

IV) $\boxed{\theta = 60^\circ}$

Kites

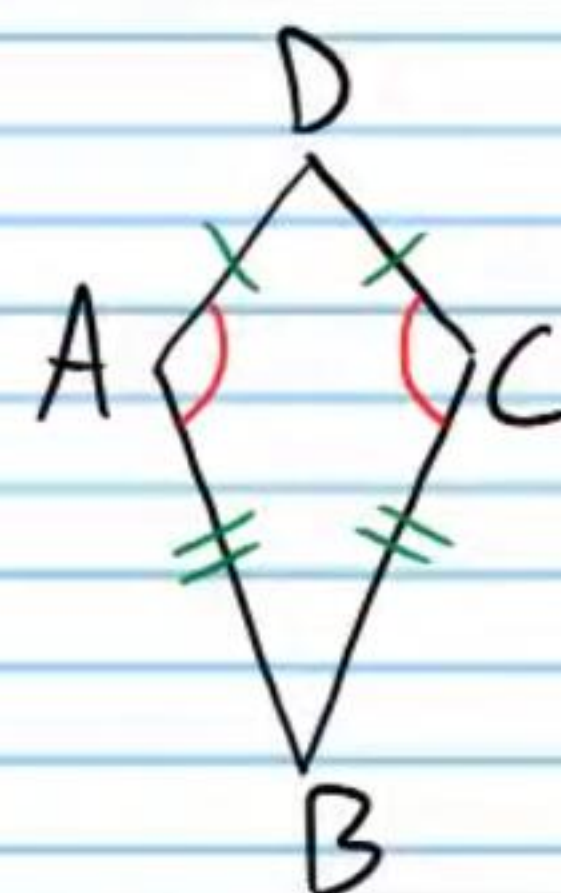
Kite Perpendicular Diagonals Theorem: If a quadrilateral is a kite, then its diagonals are perpendicular.

Example:



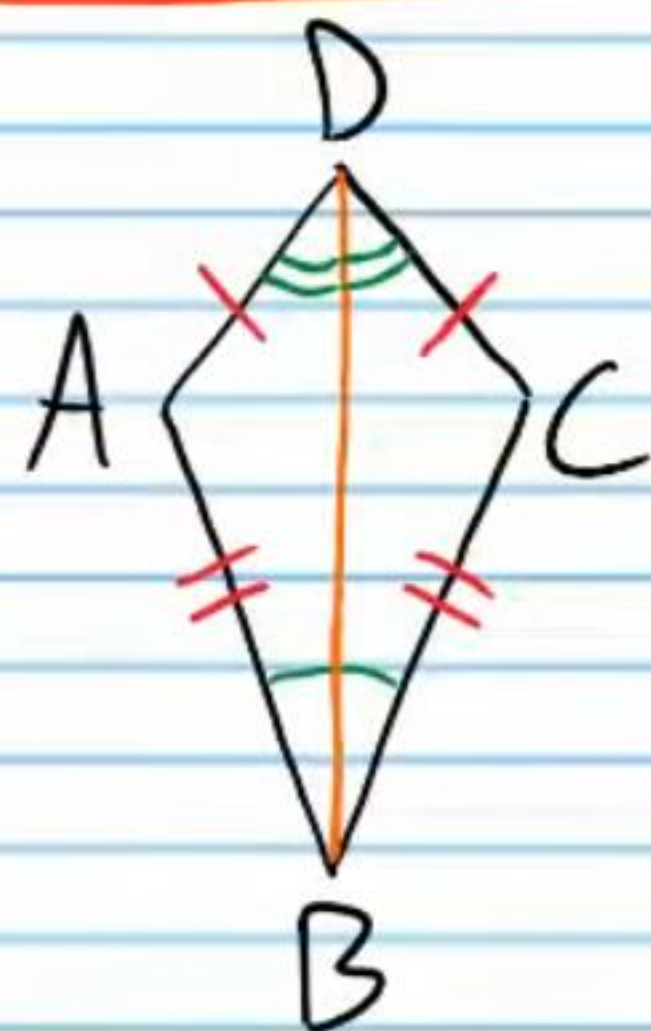
Kite Congruent Angles Theorem: If a quadrilateral is a kite, then angles between noncongruent sides are congruent.

Example:



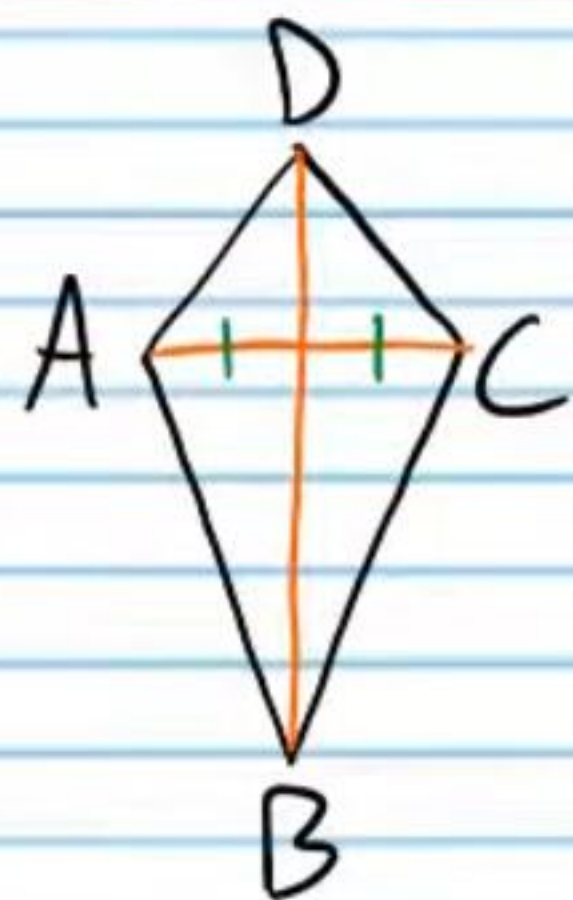
Kite Angle Bisector Theorem: The angles in a kite between congruent sides are bisected by the diagonal connecting the corresponding angle vertices.

Example:



Kite Diagonal Bisector Theorem: The diagonal with endpoints at the vertices of the angles between congruent sides bisects the other diagonal.
* Be sure to understand that only one diagonal is bisected.

Example:



Kite Right Triangles Theorem: The diagonals of a kite divide it into two pairs of congruent right triangles.

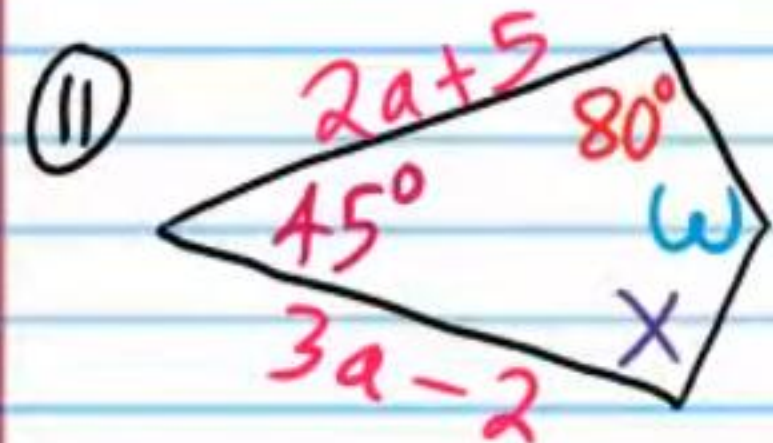
Example:



$$\triangle ADE \cong \triangle CDE$$

$$\triangle ABE \cong \triangle CBE$$

Find the unknowns in each Kite below.



I) $2a+5=3a-2$

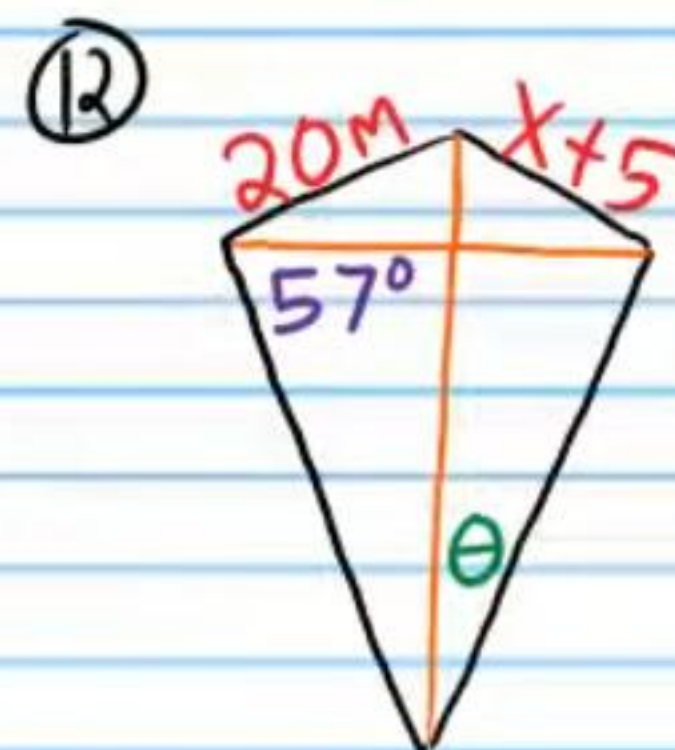
$7=a$

II) $X=80^\circ$

III) $45+80+X+W=360$

$205+W=360$

$W=155^\circ$



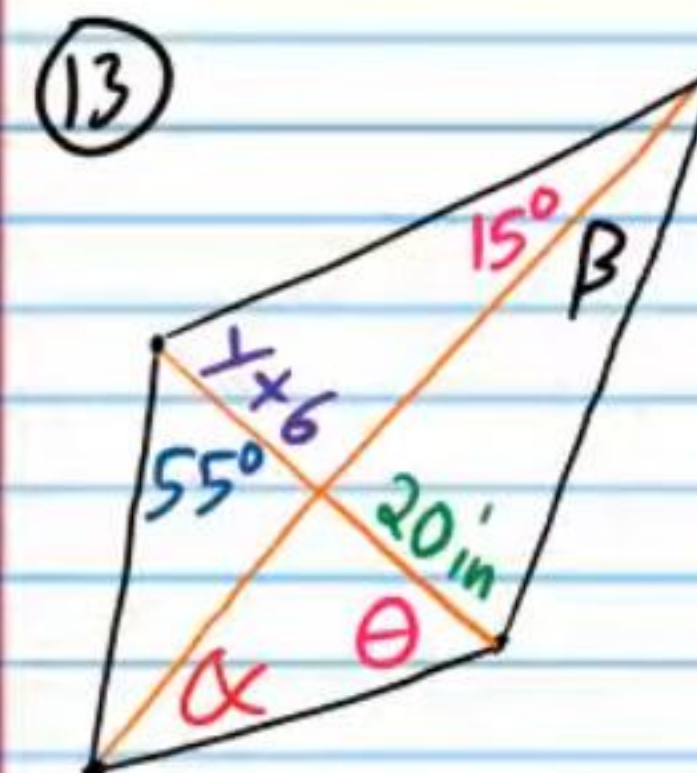
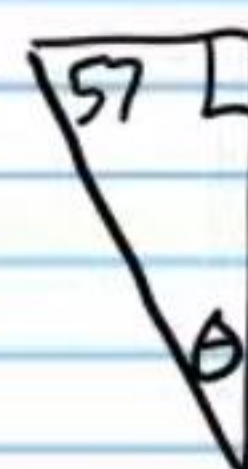
I) $20=X+5$

$15=X$

II) $57+\theta+90=180$

$147+\theta=180$

$\theta=33^\circ$

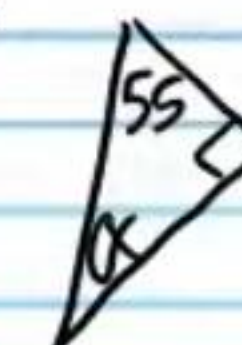


I) $Y+6=20$

$Y=14$

II) $\beta=15^\circ$

III)

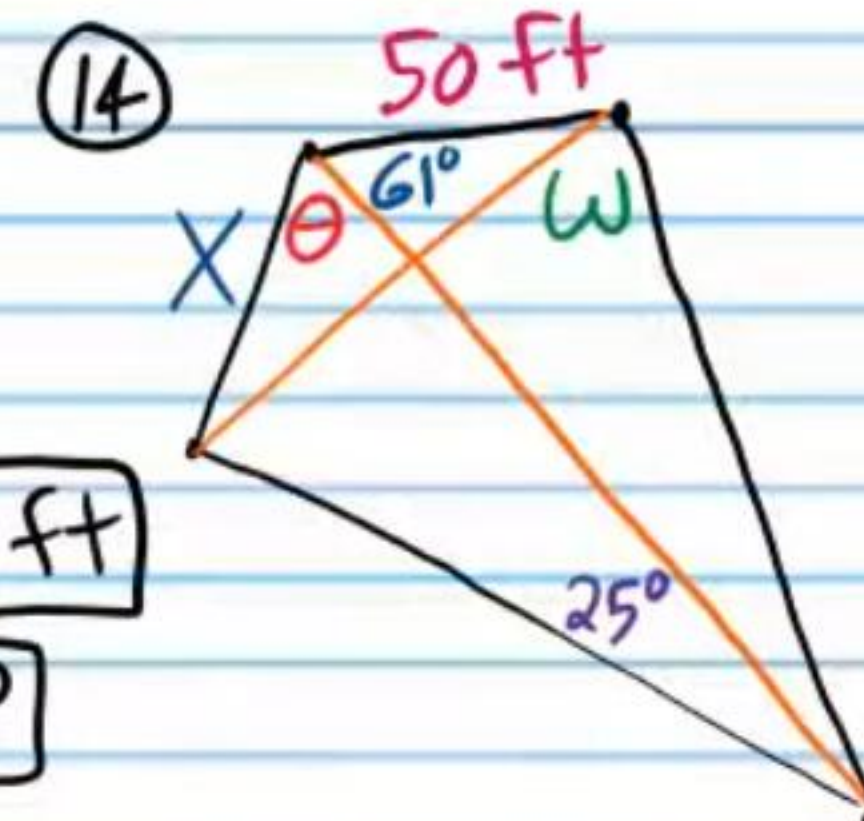


$55+\alpha+90=180$

$145+\alpha=180$

$\alpha=35^\circ$

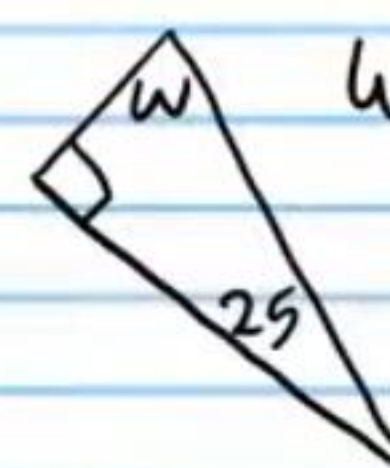
IV) $\theta=55^\circ$



I) $X=50ft$

II) $\theta=61^\circ$

III)

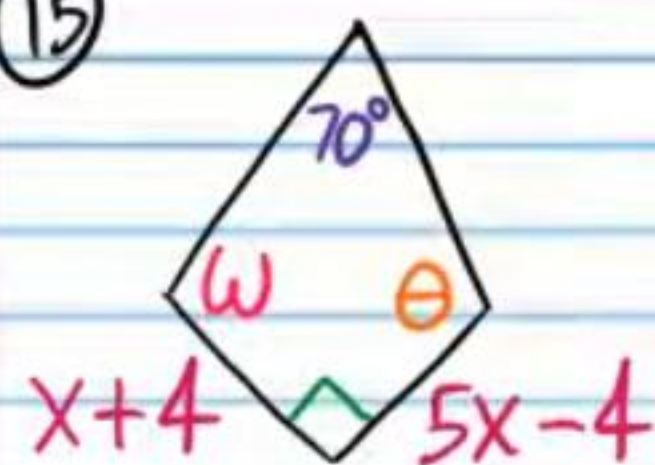


$W+25+90=180$

$W+115=180$

$W=65^\circ$

(15)



$$\begin{aligned} \text{I) } X+4 &= 5X-4 \\ 8 &= 4X \end{aligned}$$

$$\boxed{2=X}$$

$$\text{II) } W+\theta+90+70=360$$

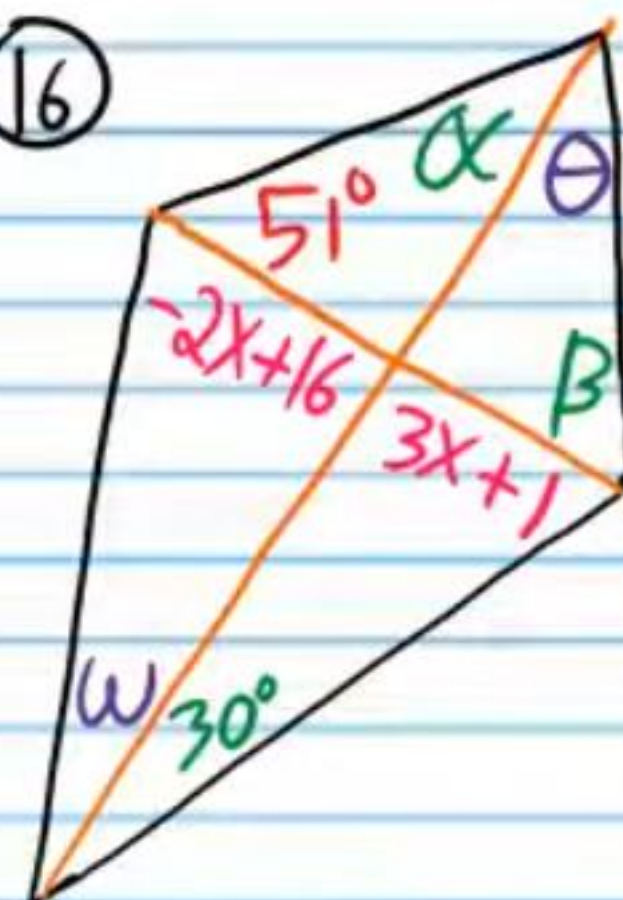
$$W+\theta+160=360$$

The same $\rightarrow W+\theta=200$

$$\boxed{W=100^\circ}$$

$$\boxed{\theta=100^\circ}$$

(16)



$$\text{I) } \boxed{W=30^\circ}$$

$$\text{II) } -2X+16=3X+1$$

$$15=5X$$

$$\boxed{3=X}$$



$$\alpha+51+90=180$$

$$\alpha+141=180$$

$$\boxed{\alpha=39^\circ}$$

$$\text{IV) } \boxed{\beta=51^\circ}$$

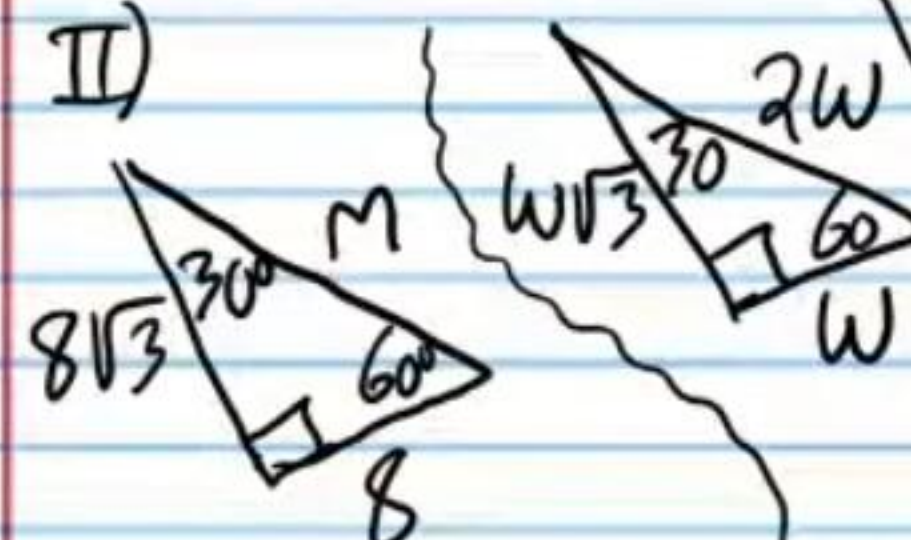
$$\text{V) } \boxed{\theta=39^\circ}$$

(17)



$$\text{I) } \boxed{L=17}$$

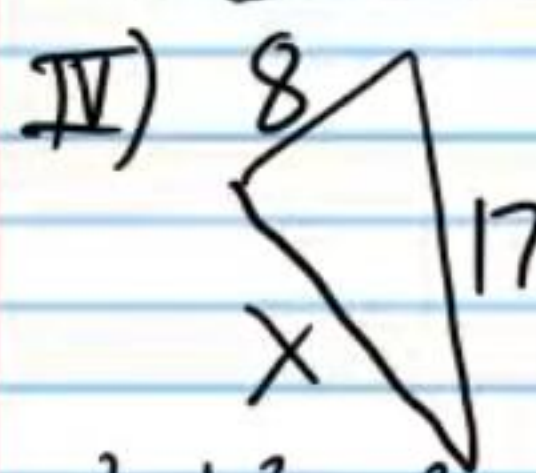
II)



$$M=2(8)=16$$

$$\boxed{M=16}$$

$$\text{III) } \boxed{Y=8}$$



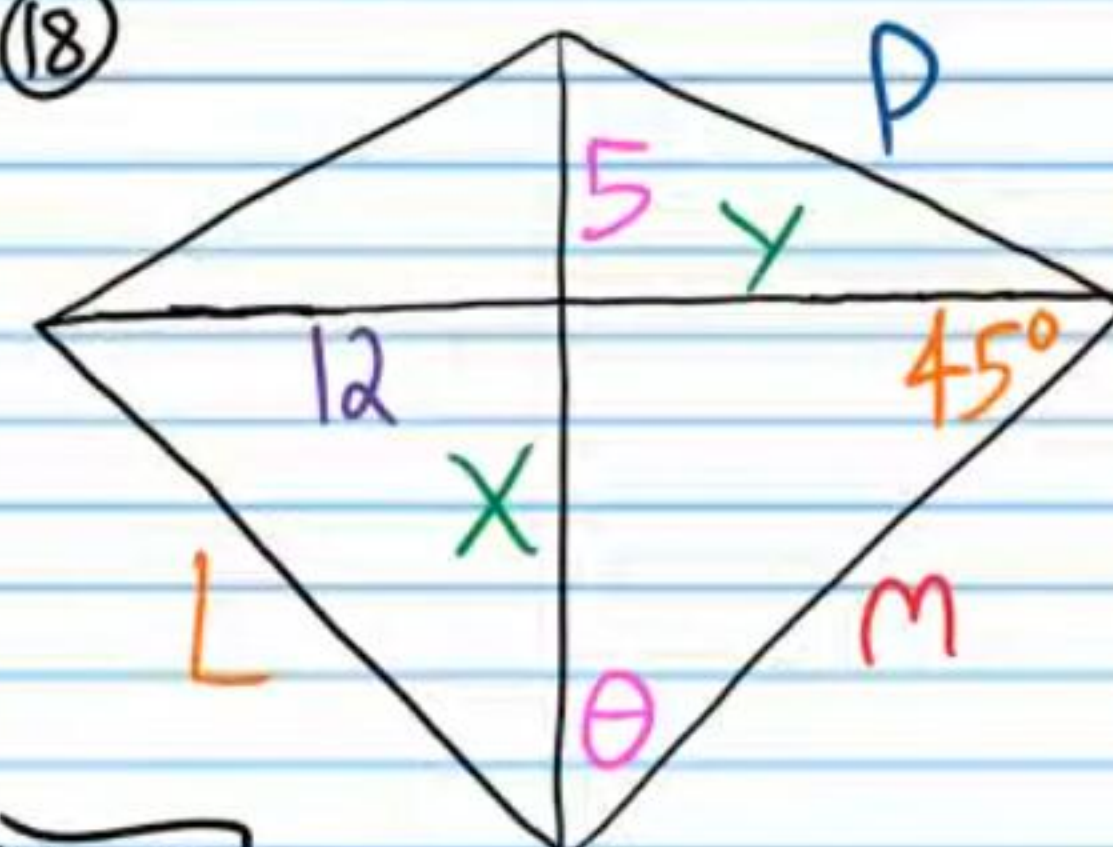
$$a^2+b^2=c^2$$

$$8^2+X^2=17^2$$

$$64+X^2=289$$

$$\text{V) } \boxed{\beta=60^\circ}$$

(18)



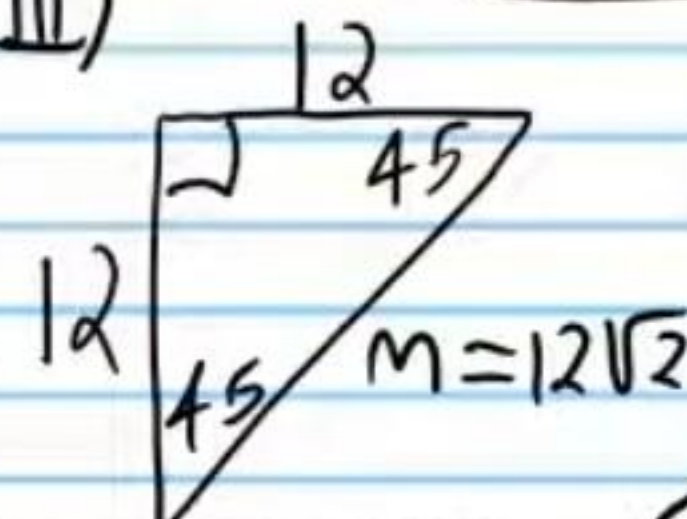
$$\text{I) } \boxed{Y=12}$$

$$\text{II) } \theta+45+90=180$$

$$\theta+135=180$$

$$\boxed{\theta=45^\circ}$$

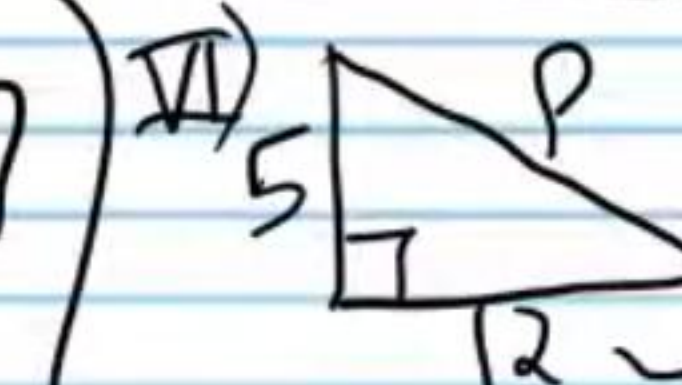
III)



$$\boxed{M=12\sqrt{2}}$$

$$\text{IV) } \boxed{X=12}$$

$$\text{V) } \boxed{L=12\sqrt{2}}$$



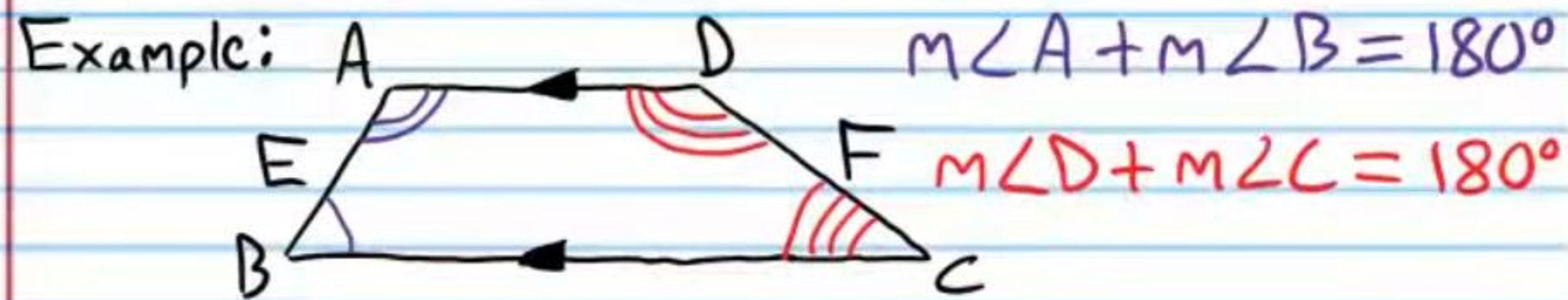
$$a^2+b^2=c^2$$

$$5^2+12^2=P^2$$

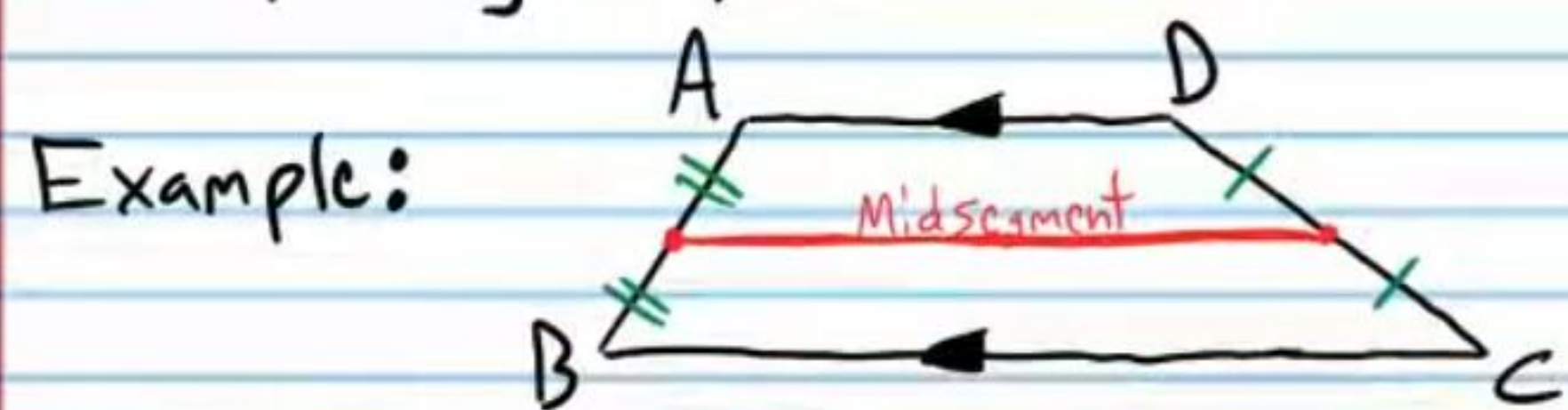
$$\boxed{13=P}$$

Trapezoids

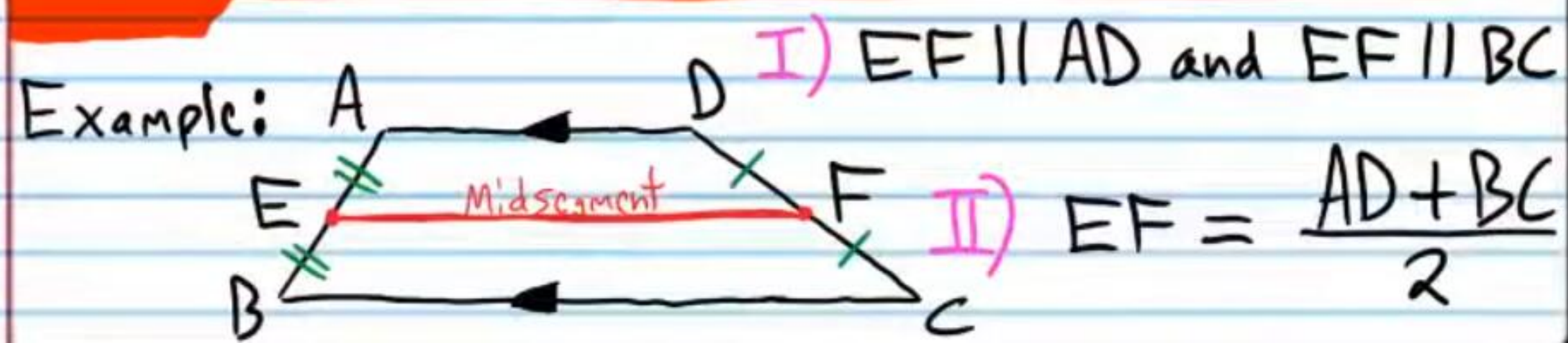
Trapezoid Non-Base Pair Theorem: The angles in non-base pairs of a trapezoid are supplementary.



Trapezoid Midsegment: A segment whose endpoints are the midpoints of the legs of the corresponding trapezoid.



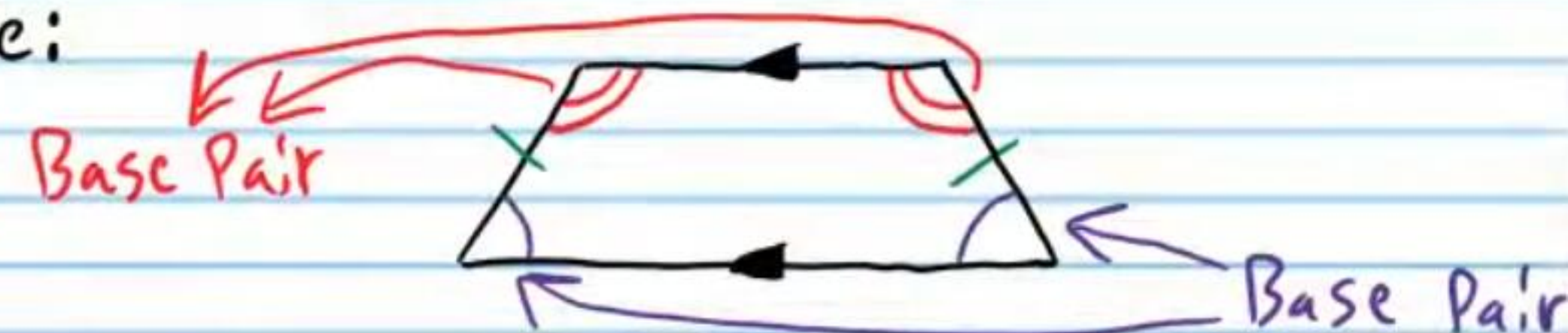
Trapezoid Midsegment Theorem: The midsegment of a trapezoid is parallel to both of the bases and its length is one half the sum of the base lengths. *Do not confuse this theorem with the Triangle Midsegment Theorem (learned in a previous class).*



Isosceles Trapezoids

Trapezoid Congruent Base Angles Theorem: If a trapezoid is isosceles, then the angles in each base pair are congruent.

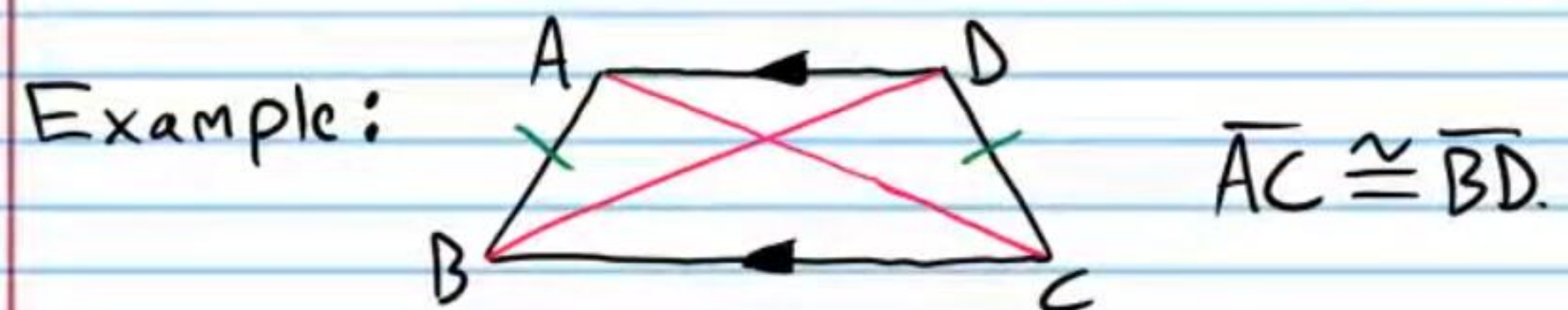
Example:



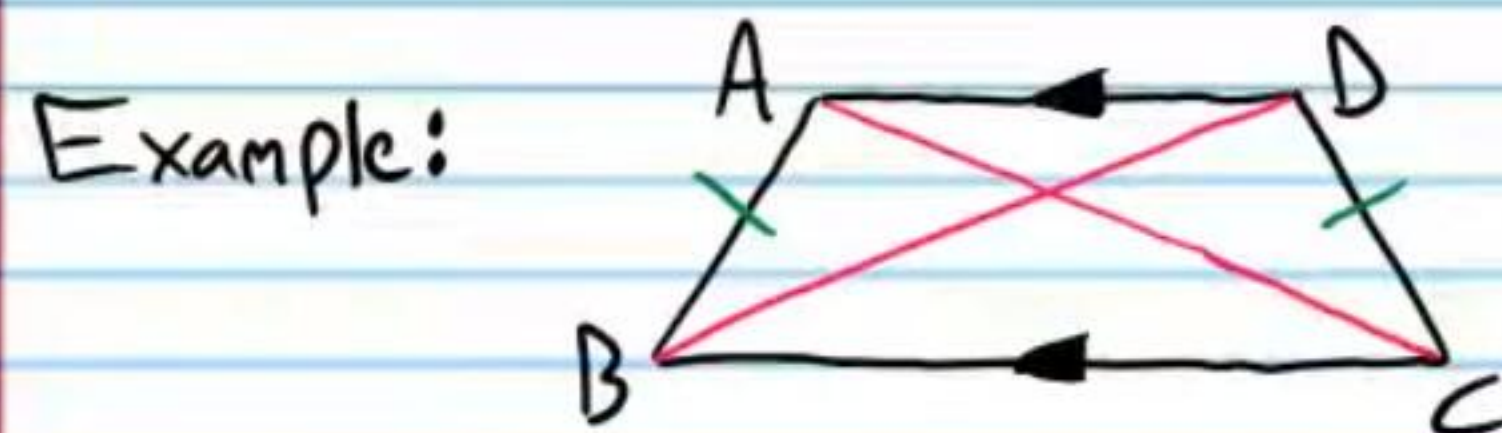
Trapezoid One Pair of Base Angles Theorem: If the angles in one base pair of a trapezoid are congruent, then the trapezoid is isosceles.



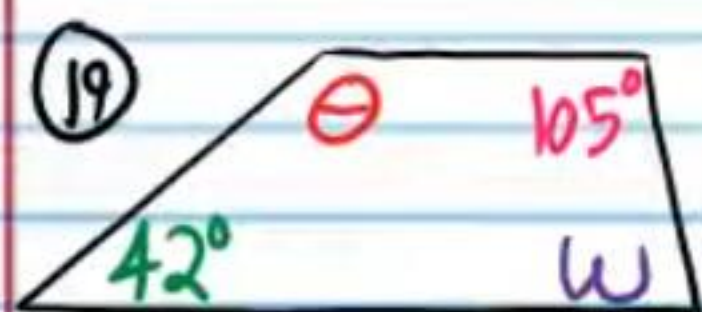
Trapezoid Diagonals Theorem: If a trapezoid is isosceles, then its diagonals are congruent.



Trapezoid Diagonals Converse Theorem: If the diagonals of a trapezoid are congruent, then it is an isosceles trapezoid.



Find the unknowns in each trapezoid below.

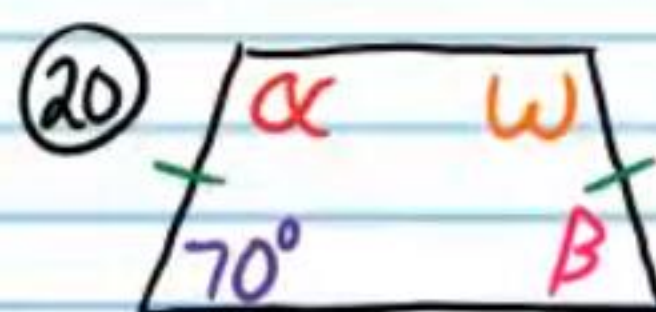


I) $\theta + 42 = 180$

$\theta = 138^\circ$

II) $105 + w = 180$

$w = 75^\circ$

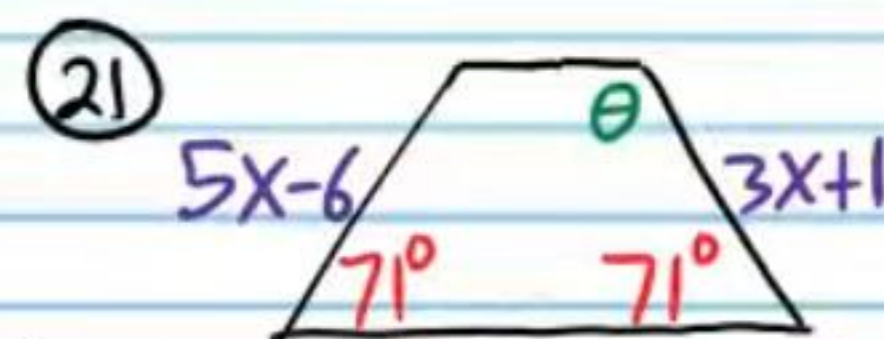


I) $B = 70^\circ$

II) $\alpha + 70 = 180$

$\alpha = 110^\circ$

III) $w = 110^\circ$



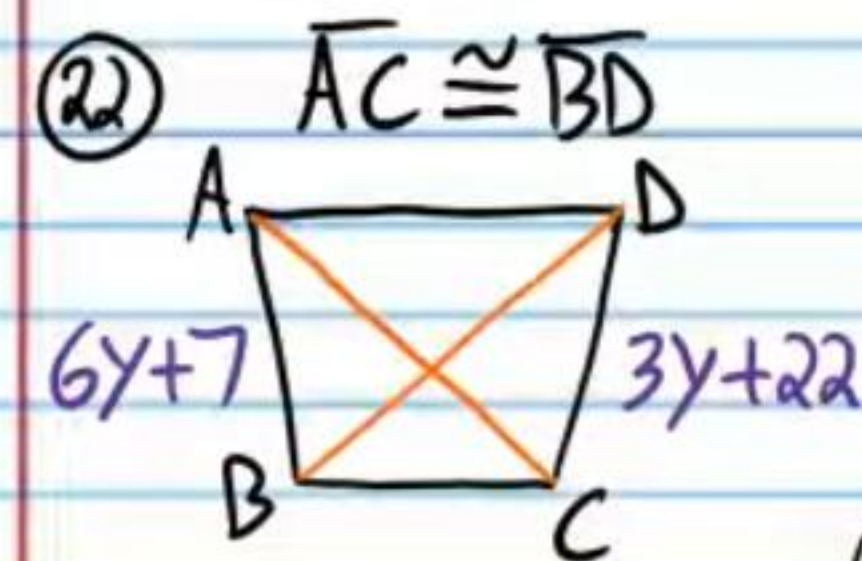
I) $5x - 6 = 3x + 1$

$2x = 7$

$x = 3.5$

II) $71 + \theta = 180$

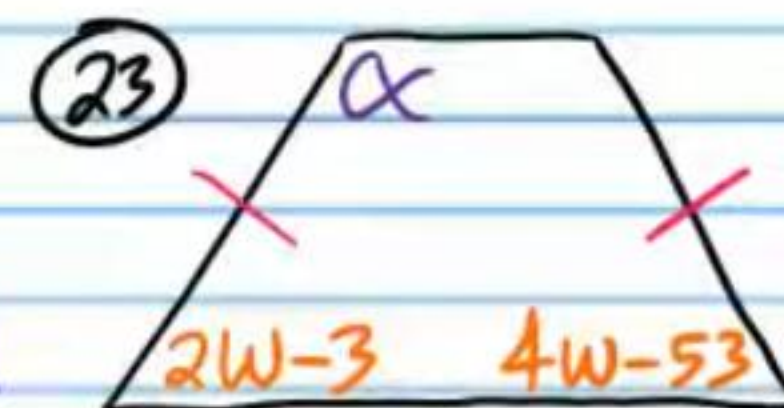
$\theta = 109^\circ$



$6y + 7 = 3y + 22$

$3y = 15$

$y = 5$



I) $2w - 3 = 4w - 53$

$50 = 2w$

$25 = w$

II) $\alpha + 2(25) - 3 = 180$

$\alpha + 47 = 180$

$\alpha = 133^\circ$

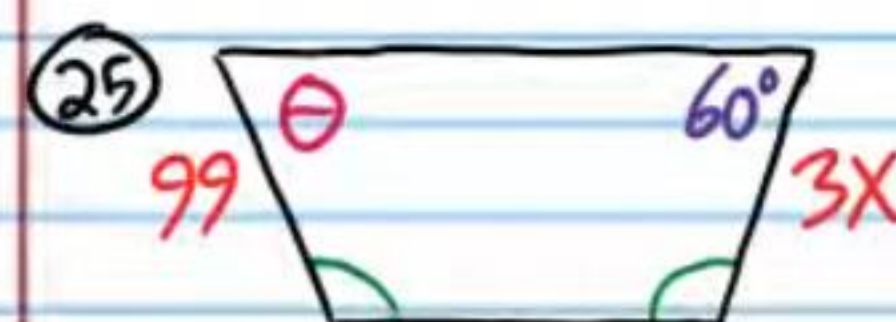
I) $\alpha + 90 = 180$

$\alpha = 90^\circ$

II) $2w + w + 30 = 180$

$3w + 30 = 180$

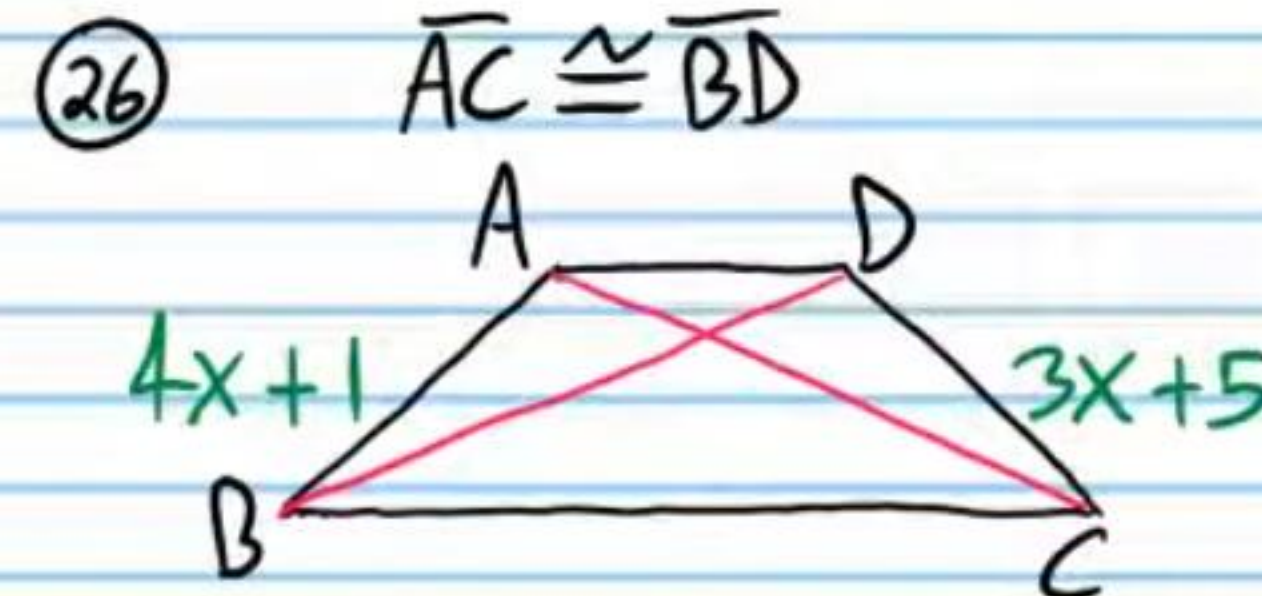
$w = 50$



I) $99 = 3x$

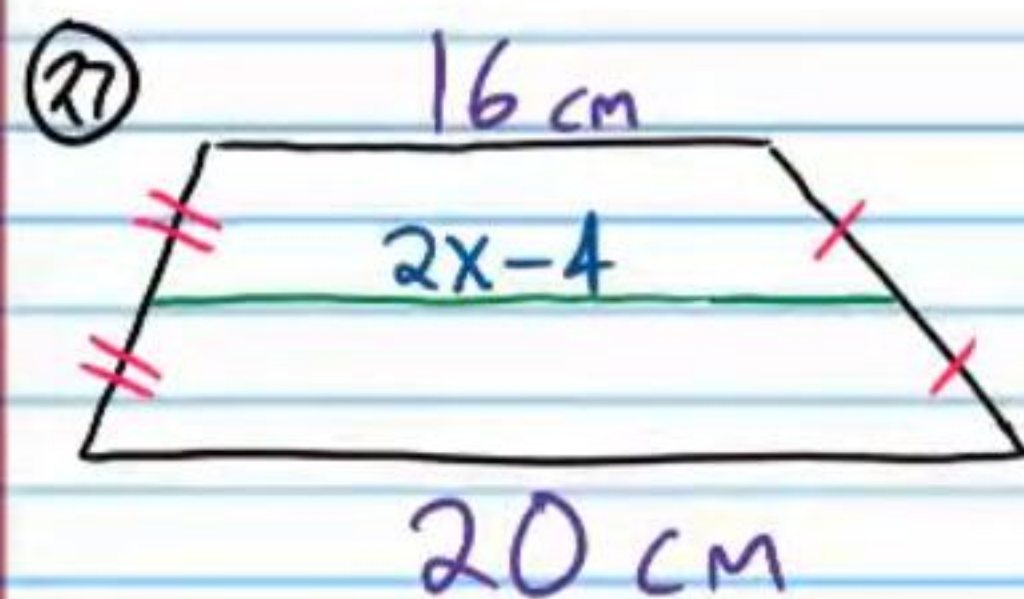
$33 = x$

II) $\theta = 60^\circ$



$4x + 1 = 3x + 5$

$x = 4$

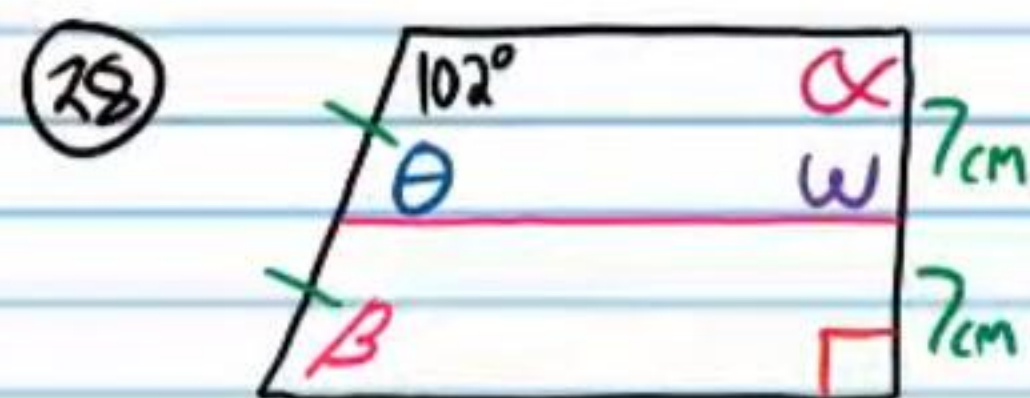


$$\frac{16+20}{2} = 2x-4$$

$$\frac{36}{2} = 2x-4$$

$$18 = 2x-4$$

$$\boxed{x=11}$$



$$\text{I) } \beta + 102 = 180$$

$$\boxed{\beta = 78^\circ}$$

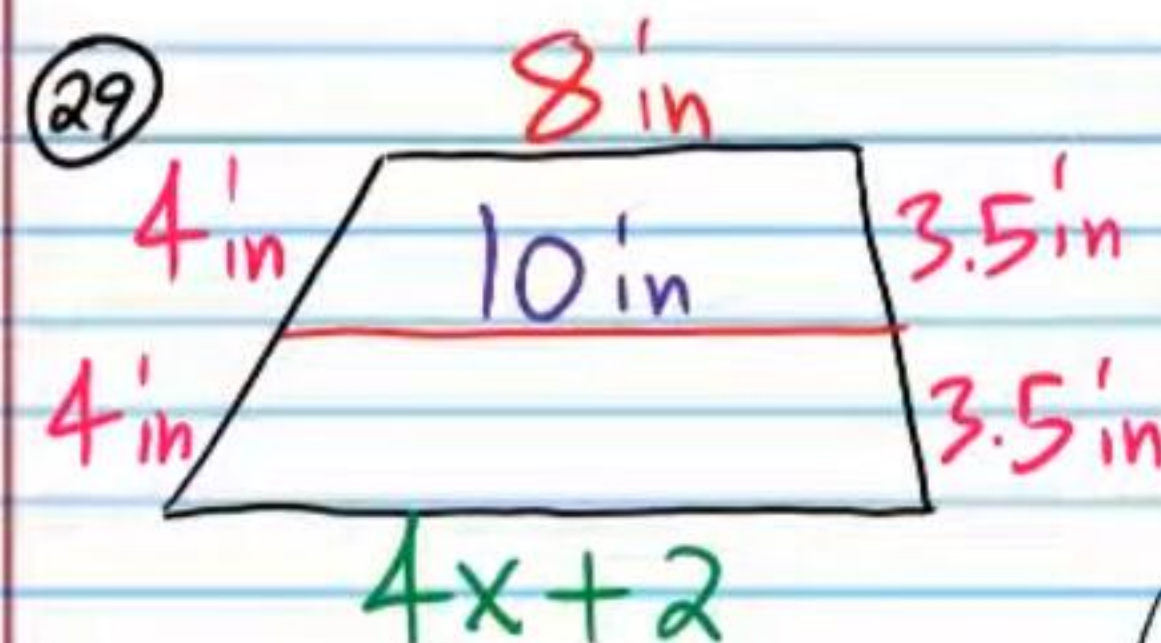
$$\text{II) } \boxed{\theta = 78^\circ}$$

$$\text{III) } \boxed{w = 90^\circ}$$

$$\text{IV) } \alpha + w = 180$$

$$\alpha + 90 = 180$$

$$\boxed{\alpha = 90^\circ}$$



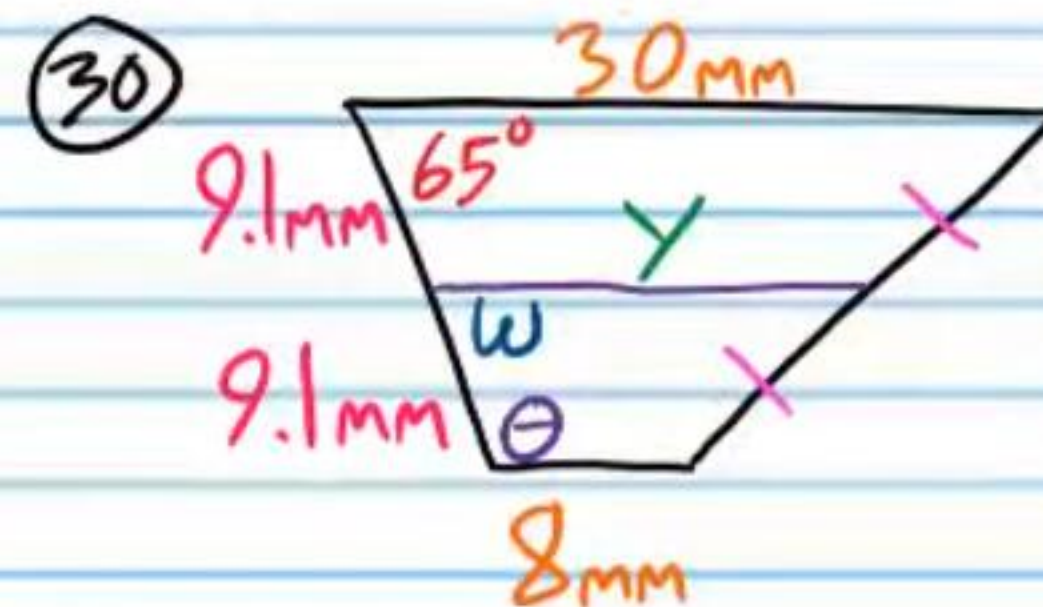
$$\frac{8+4x+2}{2} = 10$$

$$(2) \frac{8+4x+2}{2} = 10(2)$$

$$8+4x+2 = 20$$

$$10+4x = 20$$

$$\boxed{x = \frac{5}{2}}$$



$$\text{I) } y = \frac{30+8}{2}$$

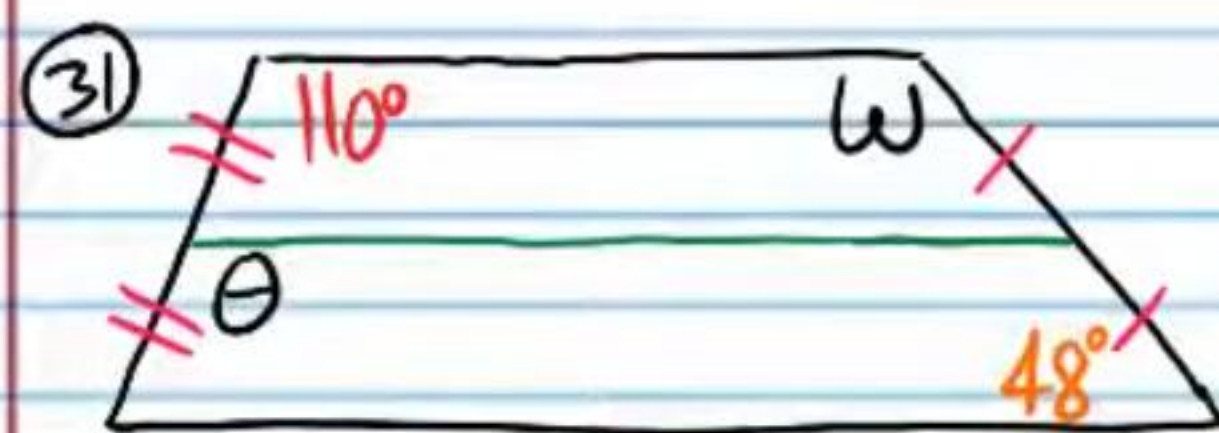
$$y = \frac{38}{2}$$

$$\boxed{y = 19 \text{ mm}}$$

$$\text{II) } \boxed{w = 65^\circ}$$

$$\text{III) } \theta + 65 = 180$$

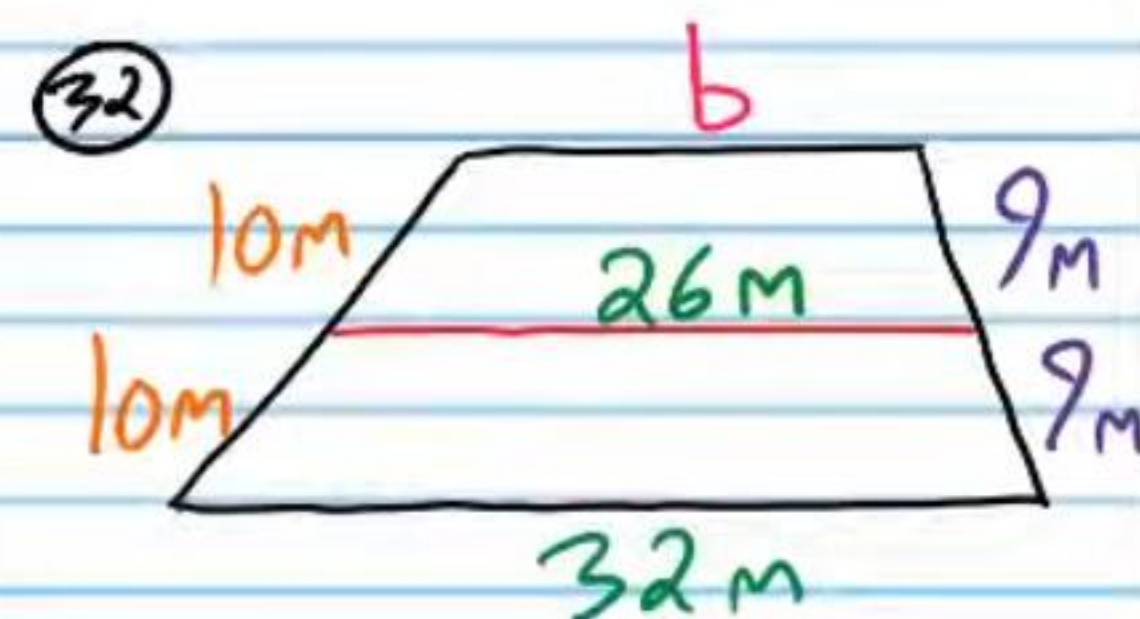
$$\boxed{\theta = 115^\circ}$$



I) $\theta = 110^\circ$

II) $w + 48 = 180$

$w = 132^\circ$



$$\frac{b+32}{2} = 26$$

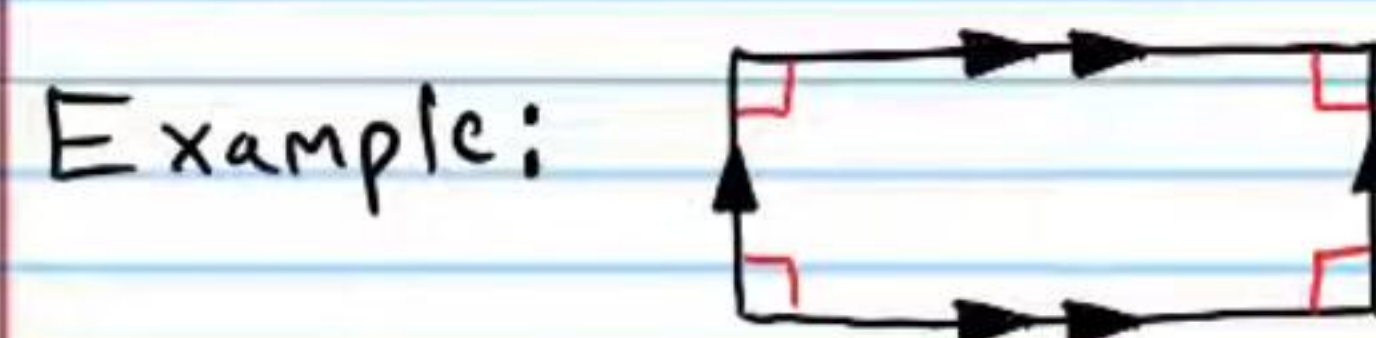
$$(2) \frac{b+32}{2} = 26 (2)$$

$$b+32 = 52$$

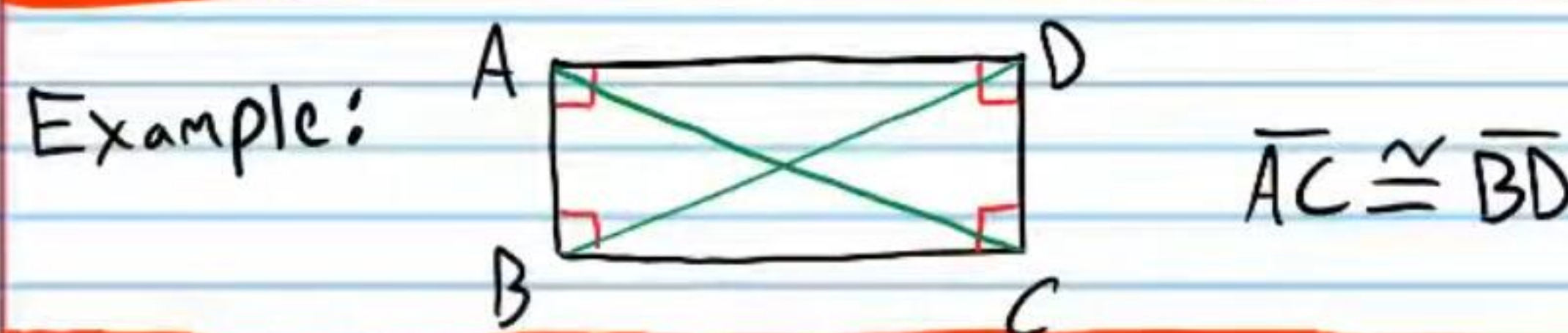
$b = 20m$

Rectangles

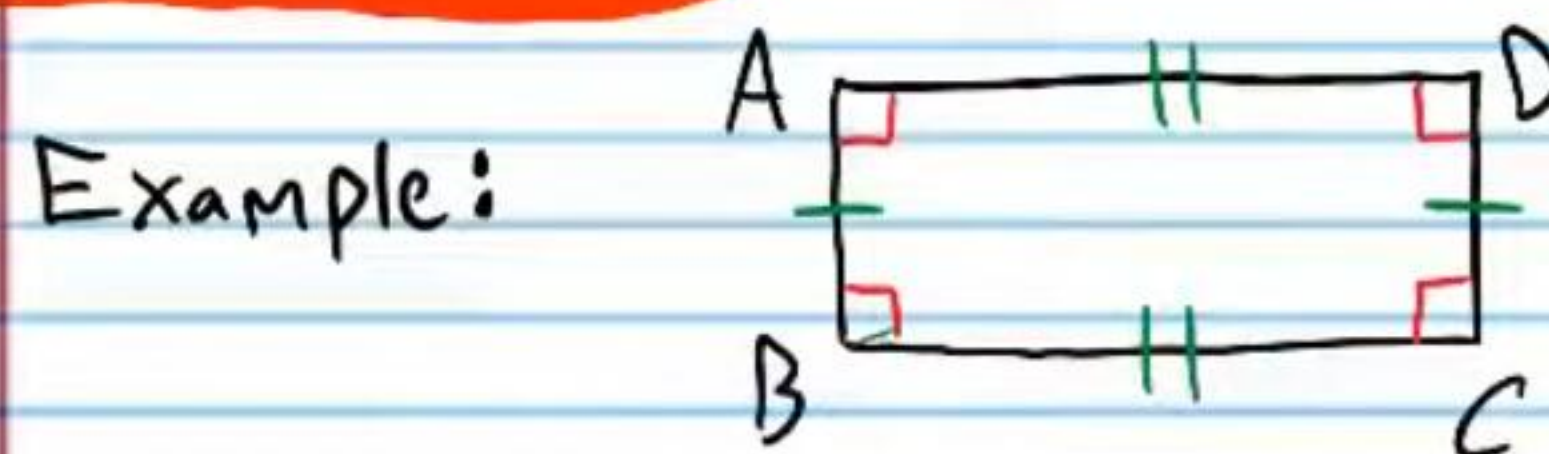
Rectangle Parallelogram Theorem: If a quadrilateral is a rectangle, then it is a parallelogram.



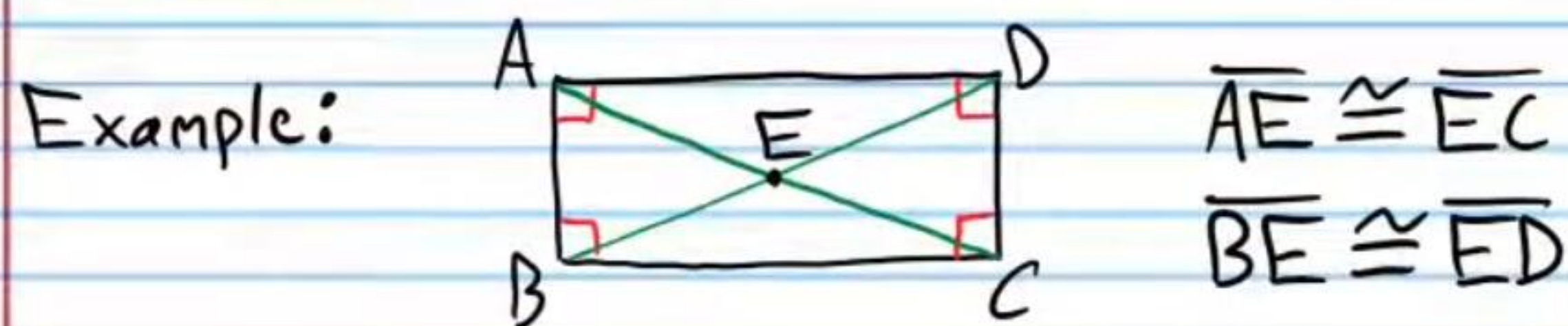
Rectangle Diagonals Theorem: If a quadrilateral is a rectangle, then its diagonals are congruent.



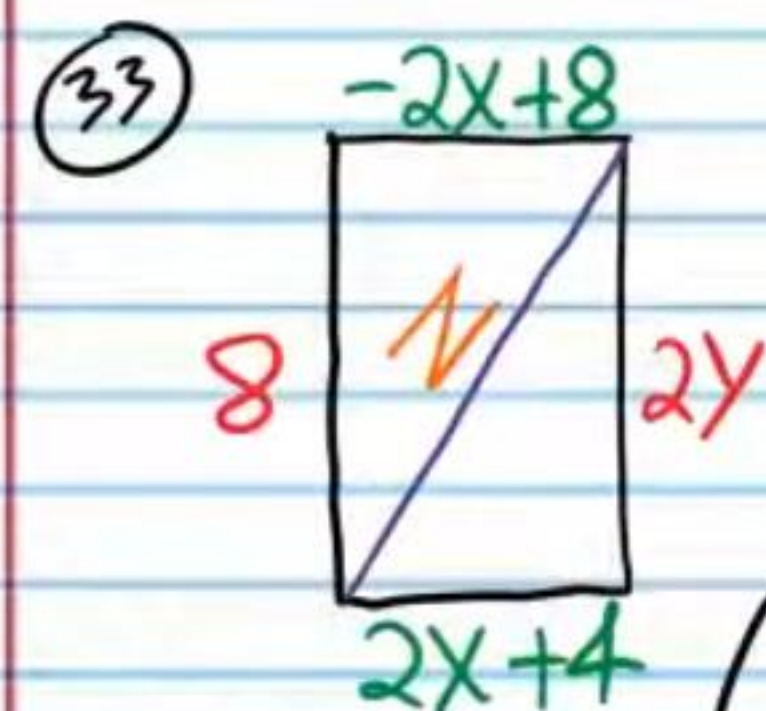
Rectangle Opposite Sides Theorem: If a quadrilateral is a rectangle, then opposite sides are congruent.



Rectangle Diagonal Bisection Theorem: If a quadrilateral is a rectangle, then its diagonals bisect each other.



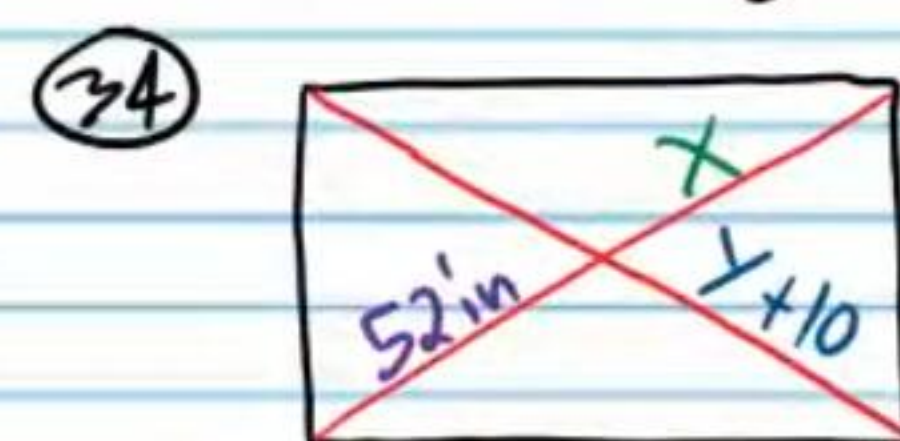
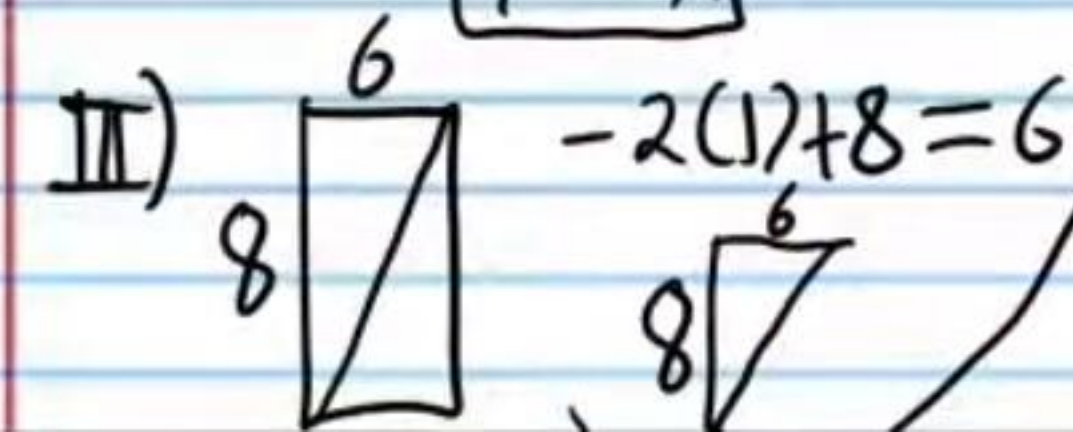
Find the unknowns in each rectangle below.



I) $8 = 2y$
 $4 = y$

II) $-2x+8 = 2x+4$ $64 + 36 = z^2$

$4 = 4x$
 $1 = x$



I) $x = 52 \text{ in}$

II) $52 = y+10$

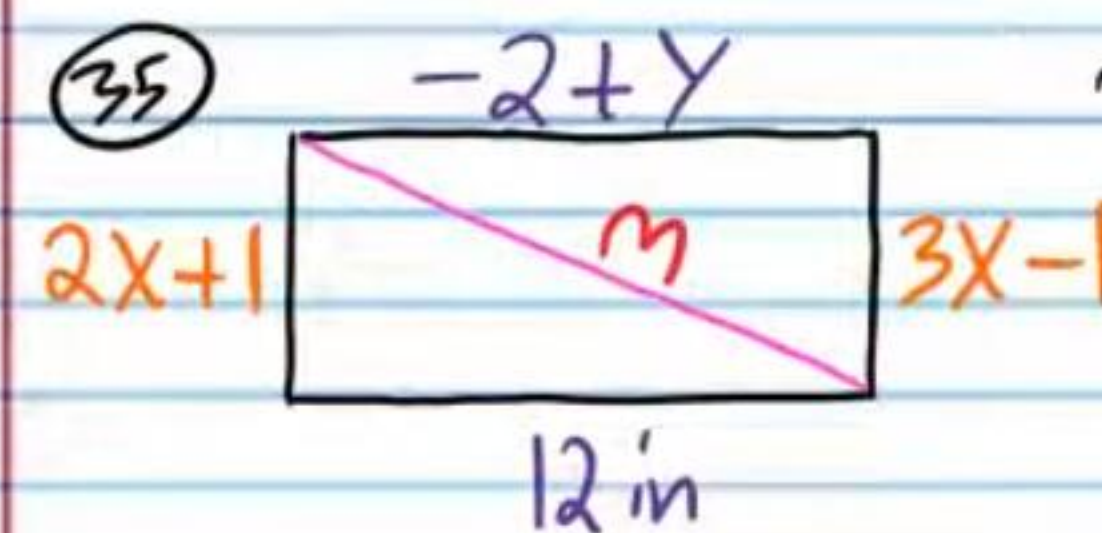
$42 = y$

$a^2 + b^2 = c^2$

$8^2 + 6^2 = z^2$

$100 = z^2$

$10 = z$

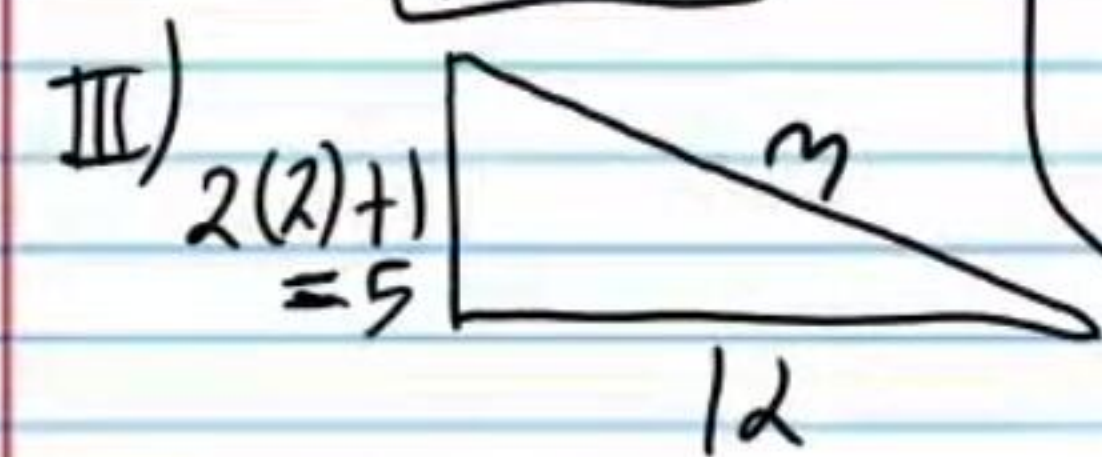


I) $2x+1 = 3x-1$

$2 = x$

II) $-2+y = 12$

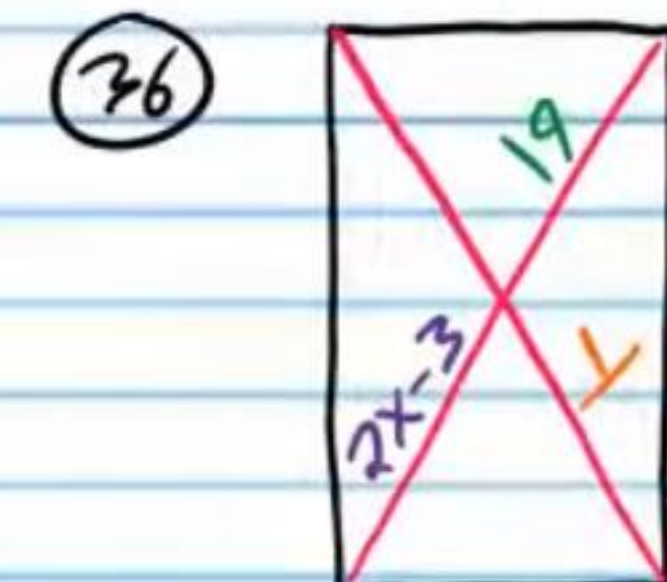
$y = 14$



$a^2 + b^2 = c^2$

$12^2 + 5^2 = m^2$

$13 = m$



I) $y = 19$

II) $19 = 2x-3$

$11 = x$

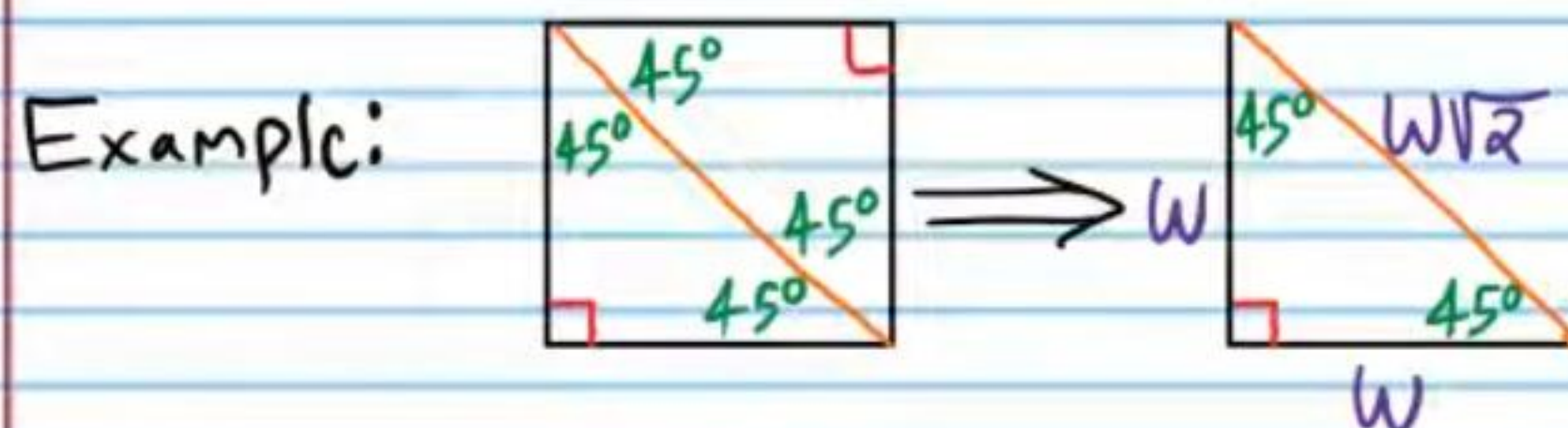
Squares

Square Parallelogram Theorem: If a quadrilateral is a square, then it is a parallelogram.

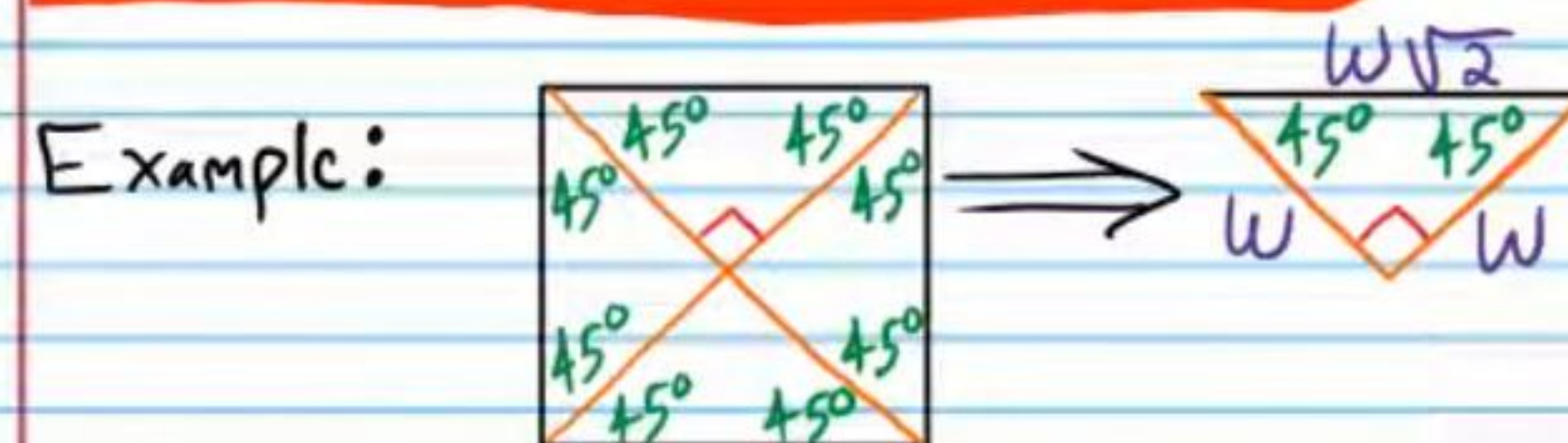
Square Rhombus Theorem: If a quadrilateral is a square, then it is a rhombus.

Square Rectangle Theorem: If a quadrilateral is a square, then it is a rectangle.

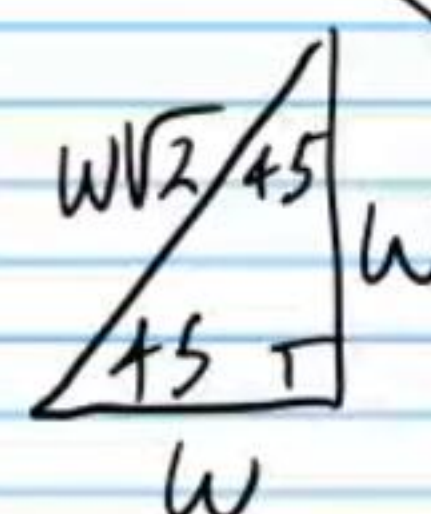
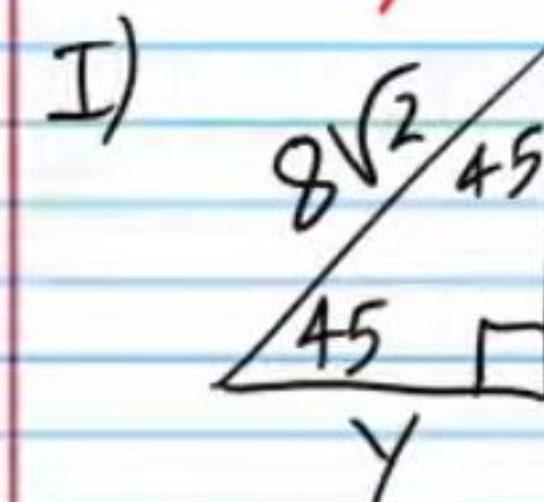
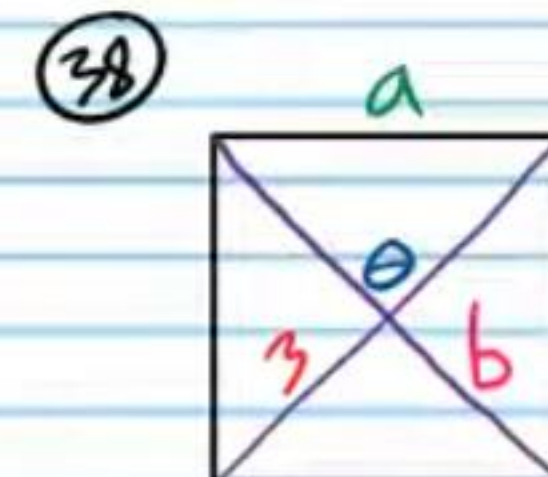
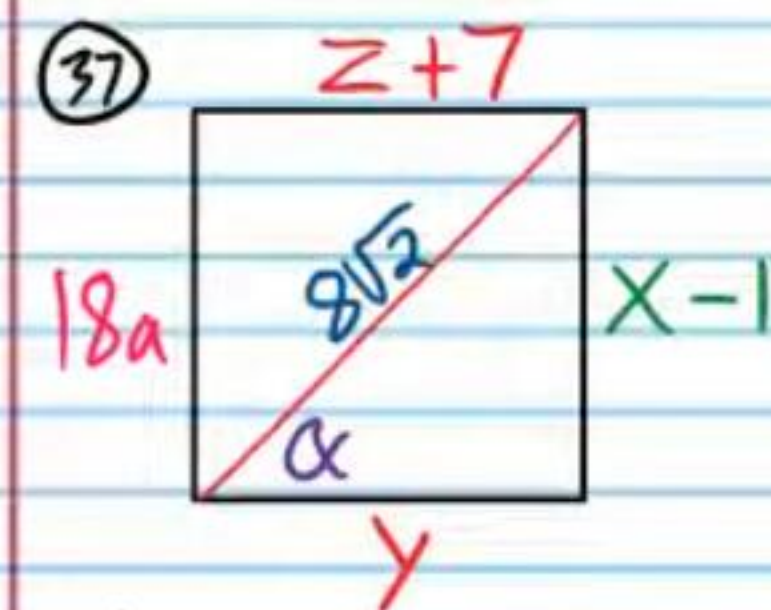
Square Two Special Triangles Theorem: If a diagonal is drawn in a square, then two 45° - 45° - 90° triangles are created.



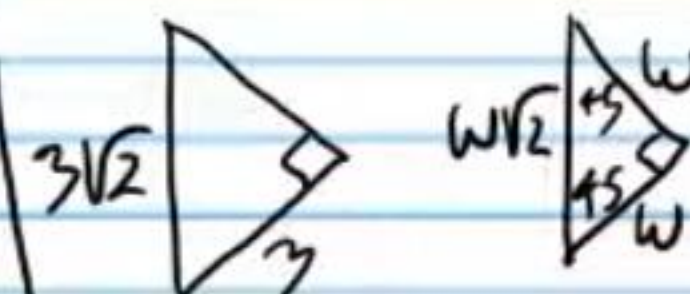
Square Four Special Triangles Theorem: If both diagonals are drawn in a square, then four 45° - 45° - 90° triangles are created.



Find the unknowns in each square below.



I) $\theta = 90^\circ$



II) $a = 3\sqrt{2}$

III) $b = 3$

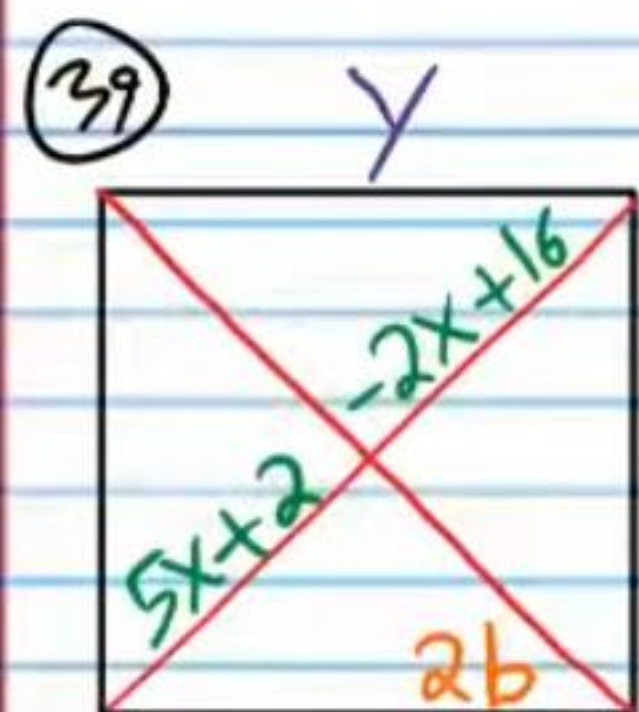
$y = 8$

II) $x - 1 = 8$
 $x = 9$

III) $18a = 8$
 $a = 8/18 = \frac{4}{9}$
 $a = \frac{4}{9}$

IV) $\alpha = 45^\circ$

V) $z + 7 = 8$
 $z = 1$



I) $2b = 45$

$b = 22.5$

II) $5x+2 = -2x+16$

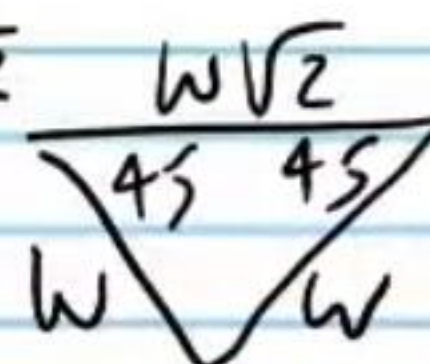
$7x = 14$

$x = 2$

III) $y = 12\sqrt{2}$

$-2(2)+16=12$

$y = 12\sqrt{2}$



I) $10 = y$

II) $10 = -a+10$

$0 = -a$

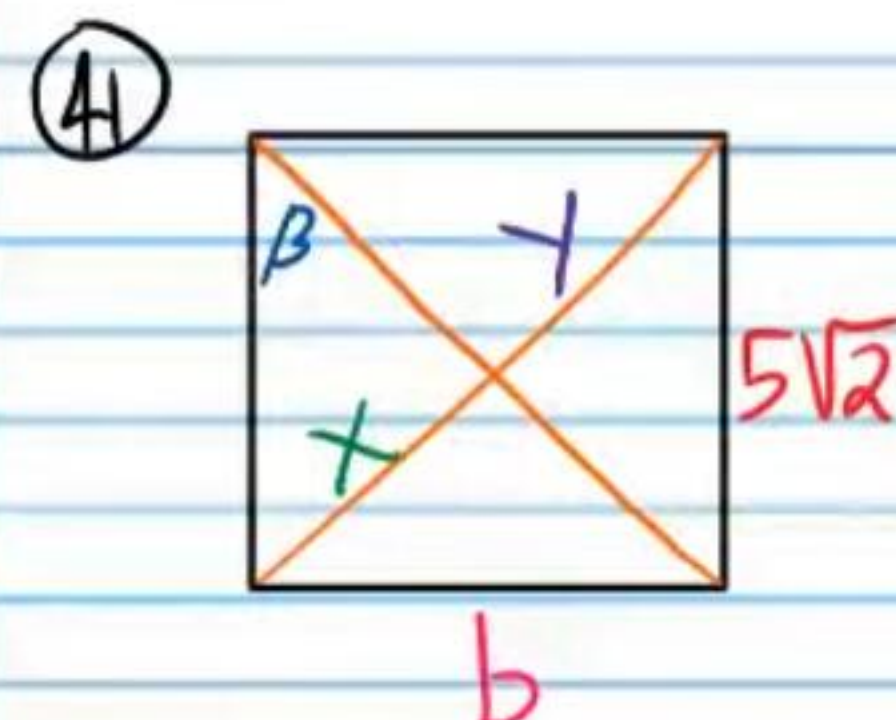
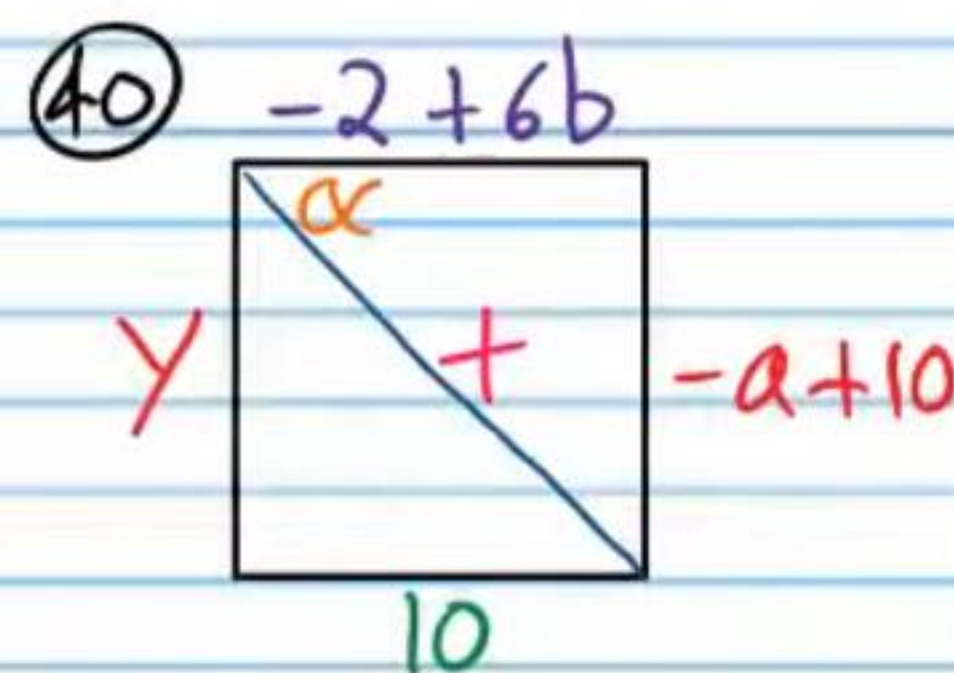
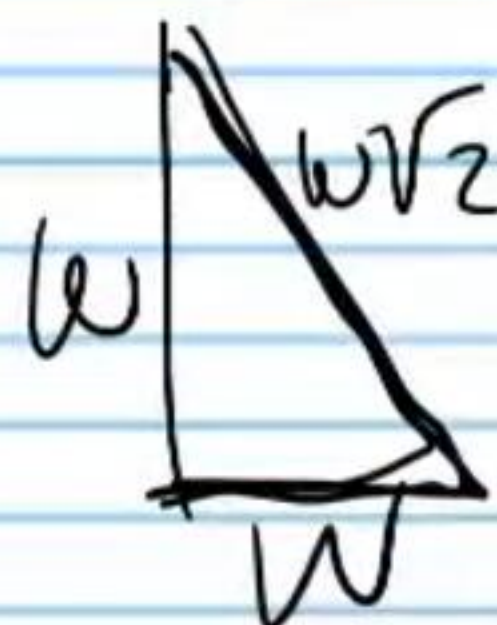
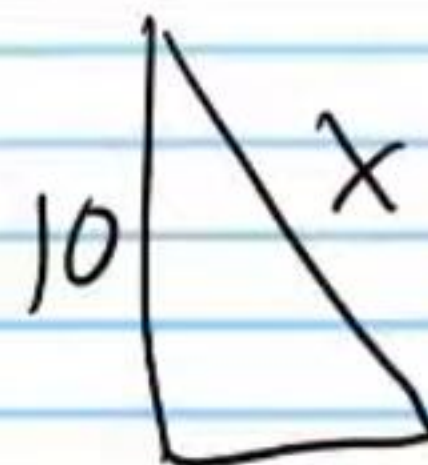
$0 = a$

III) $10 = -2+6b$

$2 = b$

IV) $\alpha = 45^\circ$

V) $x = 10\sqrt{2}$

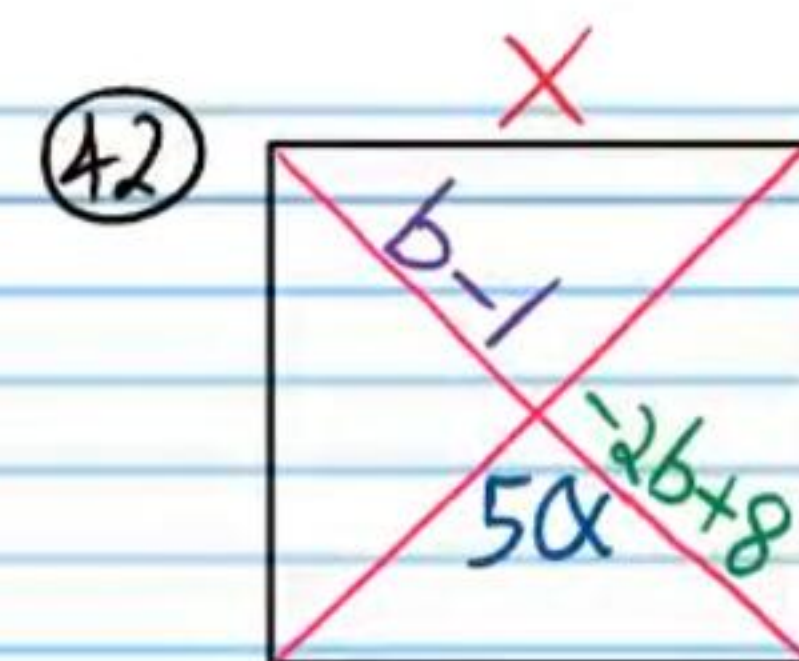
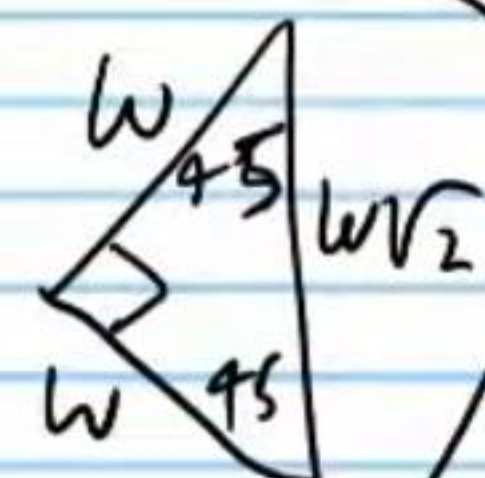
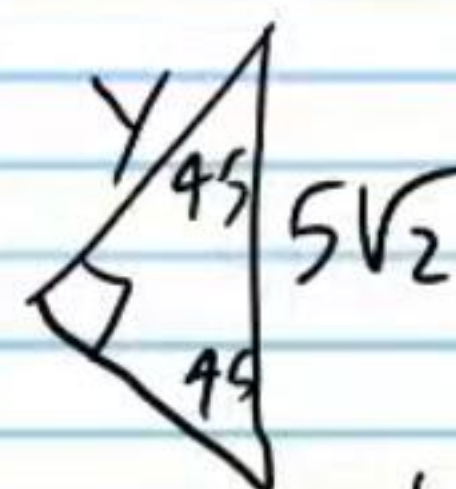


I) $\beta = 45^\circ$

II) $b = 5\sqrt{2}$

III) $y = 5$

IV) $x = 5$



I) $b-1 = -2b+8$

$3b = 9$

$b = 3$

II) $x = 2\sqrt{2}$

$x = 2\sqrt{2}$

III) $5\alpha = 90$

$\alpha = 18$

